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**Autonomous Mobile Robot
Design Documentation**

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1 Introduction

This project aims to design and develop a cost-effective autonomous mobile robot (AMR) tailored for structured indoor environments. The primary objective is to build a reliable robotic platform capable of performing predefined tasks such as material transport and workspace navigation, using practical and readily available components.

Our design incorporates essential features such as LiDAR-based obstacle detection, closed-loop stepper motor control, and a modular sensor layout to support future upgrades. Emphasis is placed on efficient power management, structured system design, and iterative hardware evaluation to meet project requirements within budget constraints.

This document presents an overview of the project's motivation, design process, component selection, and planned implementation approach.

2 Review Progress

2.1 Existing Products in the Market

To better understand the landscape of Autonomous Mobile Robots (AMRs), we studied key offerings from three leading companies: **Omron**, **MiR**, and **Geek+**. Each has distinct product lines tailored to various industrial and logistics applications. Below is a detailed breakdown and comparison.

2.1.1 Omron



Figure 1: LD Series



Figure 2: MD Series



Figure 3: HD-1500 Series

Omron offers a diverse range of AMRs under its LD, MD, and HD series. These robots are designed to operate collaboratively in dynamic environments. Omron's robots are equipped with onboard obstacle avoidance, mapping, and navigation features that are tightly integrated into the robot's core systems. The standout features include:

- **Series:** LD-60/90, MD-650/900, HD-1500
- **Payload Capacity:** 60 kg to 1500 kg
- **Navigation:** Autonomous Mapping and Navigation with LDMapEdit
- **Software:** Fleet Manager to coordinate up to 100 robots

- **Key Use Cases:** Manufacturing, Semiconductor, Healthcare
- **Safety:** ISO-certified laser scanners, speed control, and E-stop buttons

2.1.2 Mobile Industrial Robots (MiR)

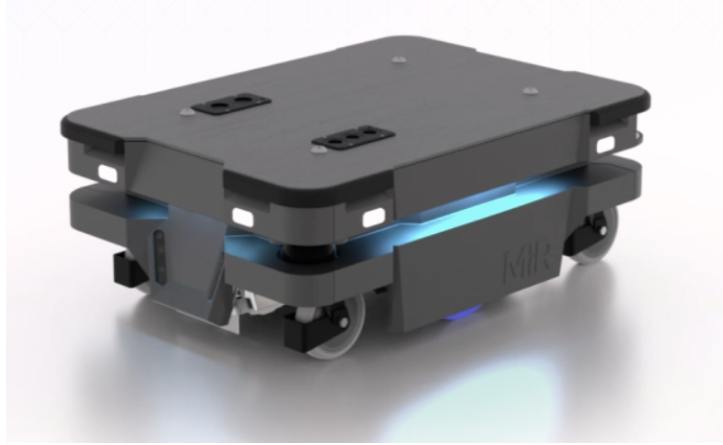


Figure 4: MiR250 AMR

MiR is a pioneer in easy-to-use AMRs, designed for rapid deployment without extensive infrastructure changes. Their robots are ideal for collaborative operations in warehouses and production lines. A strong user interface and third-party integration make MiR a flexible option.

- **Series:** MiR100, MiR250, MiR600, MiR1350
- **Payload Capacity:** 100 kg to 1350 kg
- **Navigation:** Laser-based SLAM, camera fusion
- **Software:** MiR Fleet — centralized fleet coordination
- **Key Use Cases:** Logistics, Warehouse Automation, FMCG
- **Safety:** 360° obstacle detection, dynamic path planning

2.1.3 Geek+



Figure 5: Geek+ P Series AMR

Geek+ offers a wide portfolio of AMRs, including picking, moving, and sorting robots tailored for smart warehouses. Their AI-driven platform optimizes traffic, task scheduling, and energy consumption. Geek+ is widely adopted in large-scale logistics centers.

- **Series:** P-Series (Picking), M-Series (Moving), S-Series (Sorting)
- **Payload Capacity:** 100 kg to 1200 kg+
- **Navigation:** QR code/grid-based and SLAM hybrid
- **Software:** Geek+ Robot Management System (RMS)
- **Key Use Cases:** E-commerce, Apparel, 3PL Warehousing
- **Safety:** Multi-sensor obstacle detection, AI-powered route optimization

2.1.4 Comparison of AMR Offerings

Feature	Omron	MiR	Geek+
Max Payload	1500 kg (HD-1500)	1350 kg (MiR1350)	1200+ kg (M-Series)
Navigation	Onboard laser + SLAM	Laser + Camera SLAM	QR/Grid + SLAM
Fleet Management	Omron Fleet Manager	MiR Fleet	Geek+ RMS
Ease of Use	Moderate	High (plug-and-play)	Moderate-High
Target Market	Industrial manufacturing	General logistics, hospitals	Smart warehouses, e-commerce
Safety Standards	ISO-compliant, advanced LIDARs	Full 360° sensing	AI and sensor fusion
Integration	Tight with Omron PLCs	Open API support	Customizable via SDK

Table 1: Comparison of Leading AMR Products

Conclusion: Each of these companies brings a strong value proposition to the AMR space. Omron provides tight industrial integration and reliability, MiR offers user-friendly plug-and-play solutions, and Geek+ excels in warehouse AI and logistics optimization.

2.2 Selected Reference Model

After a comprehensive evaluation of various commercially available Autonomous Mobile Robots (AMRs), we selected the **Omron LD Series**, specifically the **LD-60 platform**, as our reference design for the development of our AMR controller system.

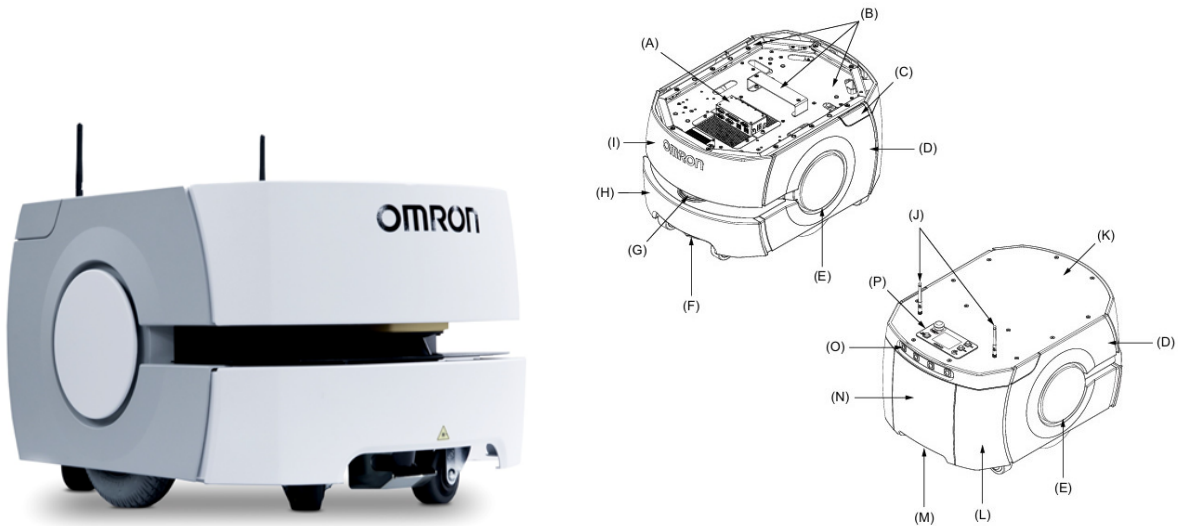


Figure 6: Omron LD-60 Reference AMR Platform

Key Features of Omron LD-60

- **Maximum Payload Capacity:** 60 kg
- **Run Time:** Up to 15 hours (no payload), approximately 12 hours (with full payload)
- **Maximum Translational Speed:** 1800 mm/s
- **Turning Radius:** 354 mm
- **Stop Position Repeatability:**
 - With CAPS: ± 12 mm, $\pm 1.0^\circ$
 - Without CAPS: up to ± 100 mm
- **Navigation:** Autonomous mapping and obstacle avoidance using LIDAR and on-board sensors
- **Footprint:** Compact, enabling navigation in narrow aisles and dynamic industrial environments

Reasons for Selection

- The LD-60 offers a well-documented and industry-proven design, making it a strong foundation for developing and benchmarking our custom controller.
- Its technical specifications align closely with our design goals in terms of speed, payload, and battery endurance.
- The platform integrates effectively with fleet management and safety systems, setting a high standard for scalability.
- CAD models, datasheets, and documentation for the LD-60 are readily available, aiding simulation and mechanical reference.

Conclusion: The Omron LD-60 platform provides a reliable and realistic reference architecture that supports our development efforts while ensuring industrial relevance and performance benchmarking.

3 Identification of Stakeholders

The stakeholder plan for our AMR controller project is developed to systematically identify, engage, and manage the various individuals and organizations involved in the design, development, deployment, and operation of the Autonomous Mobile Robot (AMR). Key stakeholders such as the Developers and Engineers, Maintenance and Support Teams, and End-users play vital roles in ensuring the system’s effectiveness, reliability, and user-friendliness.

Industries such as logistics, warehousing, and retail are expected to benefit from integrating our AMR solution into their operations. The AMR is specifically designed to automate material handling and transportation tasks, providing a significant advantage by improving efficiency, reducing manual workload, and enhancing operational safety.

Secondary stakeholders, including Regulatory Bodies, Equipment Suppliers, and Research Institutes, are also recognized for their influence on the project. Our core suppliers provide critical components like motors, sensors, microcontrollers, and batteries—elements essential to the AMR’s performance. Collaborating with specialized partners in component manufacturing and research ensures both the technical excellence and long-term sustainability of our solution.

Stakeholders were categorized into four groups—**Manage Closely**, **Keep Satisfied**, **Keep Informed**, and **Monitor**—based on their level of interest and influence. This classification helps prioritize engagement strategies and ensure effective communication throughout the development lifecycle.

Manage Closely (High interest, High Influence) End Users Businesses Interested in AMRs Developers & Engineers	Keep Informed (High interest, Low Influence) Maintenance & Support Teams Safety Analysts
Monitor (Low interest, Low Influence) Equipment Suppliers Research Institutes	Keep Satisfied (Low interest, High Influence) Investors Regulatory Bodies

Figure 7: Stakeholder Map

4 Observe Users

Understanding how users interact with Autonomous Mobile Robots (AMRs) in real-world environments is essential for designing an effective and user-friendly system. To explore user preferences and expectations, we conducted a qualitative observation study by analyzing video footage of AMR usage across different industries. The videos were reviewed with sound and subtitles turned off to sharpen our focus on physical interactions, behavioral cues, and system responses. **Videos Analyzed**

- <https://www.youtube.com/watch?v=7BenDNX5ZQU>
- <https://www.youtube.com/watch?v=LfVk2qFgPks>
- <https://www.youtube.com/watch?v=qrlx4PjfpP8>
- <https://www.youtube.com/watch?v=ESA4Cps7v2Q>
- <https://www.youtube.com/watch?v=J70Ump35T8k>
- <https://www.youtube.com/watch?v=EfBUXHnR50w>

Key Observations

- **Simplicity and familiarity:** Users preferred straightforward and familiar interfaces, with minimal manual interaction.
- **Clear feedback and updates:** Real-time notifications and system status indicators enhanced user experience and trust.
- **Autonomous navigation:** Users appreciated AMRs that could autonomously reroute around obstacles without manual input.
- **Emergency control:** Clearly accessible emergency stop buttons and fail-safe recovery mechanisms were seen as essential safety features.
- **System integration:** Compatibility with existing infrastructure, tools, and software was important for smooth operational adoption.

These insights guided several design decisions in our AMR controller, ensuring it aligns with real user needs and practical deployment scenarios.

5 Need List

After thorough evaluation and observation, we compiled a comprehensive list of user needs to ensure that the AMR controller meets the expectations of all stakeholders. These needs span across functional, safety, operational, and integration aspects, supporting the efficient and user-friendly deployment of Autonomous Mobile Robots.

- **Autonomous Navigation in Dynamic Settings:** The AMR should independently navigate and adapt to environments with moving obstacles and dynamic changes.

- **Obstacle Detection & Evasion:** Reliable sensors must detect and avoid obstacles, ensuring safe operations in crowded or shared spaces.
- **Versatile Load Handling:** Capable of transporting loads of varying sizes and weights, providing flexibility across different cargo types.
- **Long-Lasting Power with Simple Charging:** Equipped with a high-capacity battery for extended use and an efficient, user-friendly charging method.
- **Seamless System Integration:** Easy integration with existing infrastructure and management systems like warehouse software for smooth coordination.
- **User-Friendly Interface:** Simple and intuitive user interface for easy control, setup, and operation.
- **Safe Interaction with Humans & Equipment:** Designed for safe operation near humans and other machines, minimizing accident risks.
- **Quiet Operation:** Operates with minimal noise to maintain a comfortable working environment.
- **Effective Diagnostic Capabilities:** Built-in diagnostics tools to detect and report issues, enabling quick maintenance and troubleshooting.
- **Durable Build:** Constructed using tough, wear-resistant materials to enhance lifespan and reduce maintenance.
- **Safety Compliance:** Integrated safety mechanisms such as collision avoidance, emergency stop, and compliance with standards (e.g., ISO 13482).
- **Data Security & Privacy:** Ensures secure data transmission and protection, adhering to privacy and security regulations.
- **Cost-Effective Manufacturing:** Designed to be economical to produce without compromising on performance or quality.
- **Adaptable for Various Applications:** Versatile enough for deployment in different sectors including warehouses, hospitals, retail stores, and factories.
- **Remote Monitoring Capabilities:** Includes software for remote supervision and real-time system updates.
- **Energy Efficiency:** Designed for minimal power consumption to reduce operational costs and environmental impact.
- **Reliability:** High uptime and robust error-handling systems to support uninterrupted operations.
- **Scalability:** Supports coordinated operation of multiple AMRs in large-scale environments without interference or inefficiencies.

6 Component Selection

The component selection for the AMR controller robot was driven by functional requirements, performance efficiency, availability, and cost considerations. Key hardware components were evaluated and finalized after assessing multiple alternatives. Below is a summary of the selected components and their justifications:

6.1 Microcontroller Unit (MCU)

Selected: ATmega32U4

Key Characteristics:

- **Architecture:** 8-bit AVR RISC
- **Clock Speed:** Up to 16 MHz
- **Flash Memory:** 32 KB
- **SRAM:** 2.5 KB
- **EEPROM:** 1 KB
- **GPIOs:** 26 programmable I/O lines
- **Interfaces:** USB 2.0 Full Speed, SPI, I²C (TWI), USART
- **Analog Inputs:** 12-channel 10-bit ADC
- **Timers:** Four timers (two 8-bit, two 16-bit)
- **Built-in USB Support:** Enables native USB communication without external components



Figure 8: ATmega32U4 Microcontroller

Justification:

- Adequate GPIOs (> 15), USB, I2C, USART, and ADC support
- External interrupt capability
- Well-documented with strong community and SDK support
- Available development boards

Alternatives considered: SAM E70, STM32 F-Series, STM32 H-Series, RP2040

6.2 Single-Board Computer (SBC)

Selected: Jetson Nano

Key Characteristics:

- Quad-core ARM Cortex-A57 @ 1.43 GHz CPU
- 128-core Maxwell GPU
- 4 GB LPDDR4 memory
- Rich I/O (40-pin GPIO, I²C, SPI, UART)
- Excellent community and development ecosystem via NVIDIA JetPack



Figure 9: Jetson Nano Development Board

Alternatives considered: Jetson AGX Orin, AMD Ryzen V1000, Intel NVC

6.3 LIDAR Selection

Autonomous navigation and obstacle avoidance are core functionalities in any AMR system. Therefore, selecting an appropriate LIDAR sensor is crucial for ensuring precise environmental perception. Key factors considered in this selection include:



Figure 10: Lakibeam 1S LIDAR Sensor

- Measurement range (minimum and maximum distance)
- Accuracy and angular resolution
- Field of view (angle of detection)
- Sampling rate
- Communication interface
- Dimensional capability (2D or 3D)
- Cost and ease of integration

Comparison of Available LIDAR Sensors

LIDAR	Price (USD)	2D/3D	Min Dist. (m)	Max Dist. (m)	Accuracy Score	Angle	Sampling Rate	Interface
LakiBeam 1S	310	2D	0.1	15	9	270°	18000	Ethernet UDP
YDLIDAR T-mini Pro	72	2D	0.12	10	6	360°	5000	UART
STL-19P D500	96	2D	0.03	12	7	360°	5000	UART
Velodyne VLP-16	499	3D	0.1	100	8	360°	300000	Ethernet
LS01B LIDAR Scanner	378	2D	0.1	16	8	360°	14000	UART
RPLIDAR S2P	400	2D	0.2	50	9	360°	18000	UART
VF 2D 360° LIDAR	499	2D	0.1	25	8	360°	20000	UART

Table 2: Comparison of Selected LIDAR Modules

Selected LIDAR Module

After evaluating the listed options based on technical specifications, performance metrics, and interface compatibility, we selected the **Lakibeam 1S** as the LIDAR sensor for our AMR platform.

- **Accuracy:** High, with an accuracy score of 9
- **Field of View:** 270° — ideal for forward-facing obstacle detection
- **Sampling Rate:** 18000 — ensures dense and fast environmental scanning
- **Interface:** Ethernet UDP — fast and robust data communication
- **Min/Max Range:** 0.1 m to 15 m — suitable for indoor dynamic environments
- **Dimensionality:** 2D — sufficient for horizontal floor navigation

Cost of Selected Module: \$310 (approximately Rs. 90,600)

The Lakibeam 1S offers a strong balance between cost and performance, and integrates well with our Jetson Nano SBC via Ethernet UDP for high-speed data transmission and real-time processing.

6.4 Motor Selection

6.4.1 Estimated Weight

- Motors = $3.6 \text{ kg} \times 2 = 7.2 \text{ kg}$
- Gears = $0.92 \text{ kg} \times 2 = 1.84 \text{ kg}$
- Payload = 5 kg (requirement)
- Battery < 3 kg (assumption)
- Lidar $\approx 0.4 \text{ kg}$
- Chassis + Enclosure = 5 kg (estimated)
- Jetson Nano + peripherals = 2 kg

$$\text{Total Estimated Weight} \approx 24.44 \text{ kg} \Rightarrow \text{Weight} \approx \boxed{25 \text{ kg}}$$

6.4.2 Minimum Displacement

We consider a wheel diameter of 15 cm (reference design diameter = 20 m). The stepper motor under consideration is the NEMA 24, commonly used in robotics for its balance between torque and precision.

NEMA 24 Stepper Motor Specifications:

- Standard step angle: 1.8° per full step (from manufacturer datasheet)
- Full rotation: $360^\circ = 200 \text{ steps}$
- We chose not to micro-step as it significantly reduces the motor torque output

Minimum Displacement Analysis:

- Without microstepping, minimum angular displacement:

$$\frac{1.8^\circ}{360^\circ} \times 15 \text{ mm/rev} = 0.075 \text{ mm/step}$$

- Resulting linear displacement: $\approx 2.4 \text{ mm}$ per step

LD Series Comparison:

- Assumed stop position repeatability: 10 mm (from datasheet specifications)
- Feasibility assessment: Our motor's minimum displacement capability (2.4 mm per step) is within the required spec
- **Conclusion:** Standard full-stepping is sufficient to achieve the required precision

6.4.3 Torque and Gear Ratio

TORQUE CURVE AT 48V

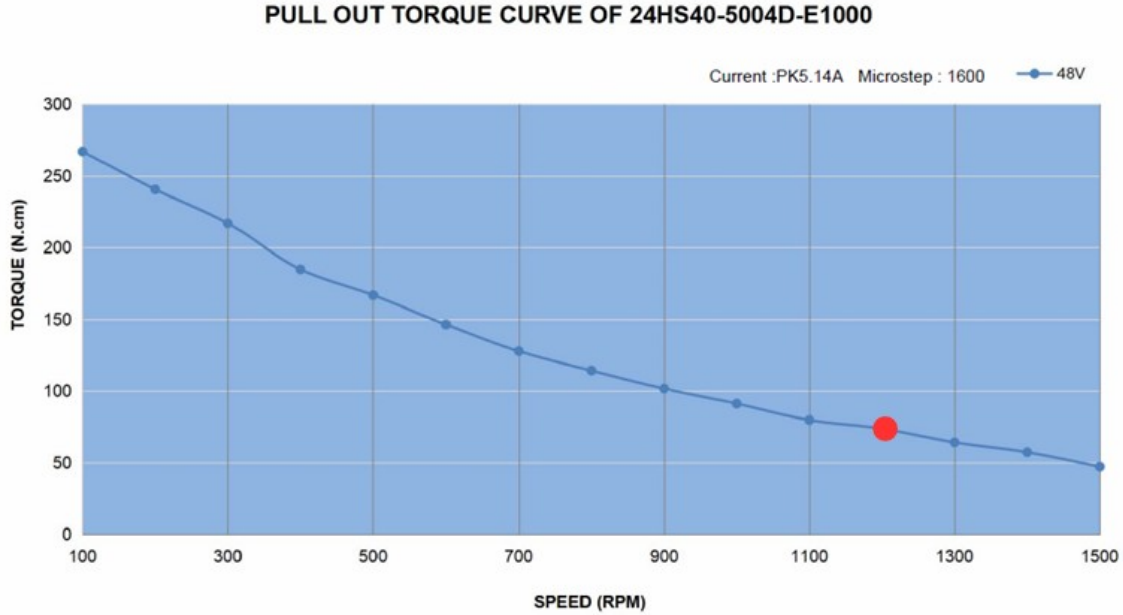


Figure 11: Torque Curve at 48 V

- Gear ratio: 10:1
- To achieve 120 RPM at the wheel, motor must rotate at 1200 RPM
- From the torque curve of NEMA 24 at 48 V and 1200 RPM:

$$\text{Motor torque} \approx 0.75 \text{ Nm}$$

- Output torque after gearing:

$$0.75 \times 10 = 7.5 \text{ Nm}$$

- Considering losses due to friction and gear inefficiencies, conservatively assumed:

$$\boxed{\text{Usable Output Torque} \approx 6.0 \text{ Nm}}$$

6.4.4 Velocity and Acceleration

We compare the torque-limited drive force to the traction limit and take the smaller to find the true maximum acceleration.

$$r = 0.075 \text{ m}, \quad T_{\text{wheel}} = 6.0 \text{ Nm}$$
$$F_{\text{drive}} = 2 \times \frac{T_{\text{wheel}}}{r} = 2 \times \frac{6.0}{0.075} = 160 \text{ N}$$

$$a_{\text{torque}} = \frac{F_{\text{drive}}}{m} = \frac{160}{25} = 6.4 \text{ m/s}^2$$

$$F_{\text{traction,max}} = \mu m g = 0.6 \times 25 \times 9.8 = 147 \text{ N}$$

$$a_{\text{traction}} = \frac{F_{\text{traction,max}}}{m} = \frac{147}{25} \approx 5.88 \text{ m/s}^2$$

$$a_{\text{max}} = \min(a_{\text{torque}}, a_{\text{traction}}) = 5.88 \text{ m/s}^2$$

Since $5.88 \text{ m/s}^2 \gg 0.6 \text{ m/s}^2$, the acceleration target is readily achieved without wheel slip.

6.4.5 Selected Motor Summary

After evaluating motion requirements, weight, torque, precision, and cost constraints, the selected motor for the AMR platform is:

- **Motor Type:** NEMA 24 Closed-Loop Stepper Motor
- **Holding Torque:** 4.0 Nm
- **Operating Voltage:** 24 V to 48 V
- **Driver Type:** Integrated closed-loop driver with encoder feedback
- **Gearbox:** Planetary Gearbox with 10:1 reduction ratio
- **Output Torque (after gearbox):** Approx. 6.0 Nm (conservatively estimated)
- **Weight per motor:** 3.6 kg
- **Gearbox weight:** 0.92 kg



Figure 12: NEMA 24 Closed-Loop Stepper Motor

Reasons for Selection:

- Provides sufficient torque with gear reduction to drive the estimated 25 kg robot with margin.
- Full-step resolution offers displacement accuracy within the required stop position tolerance.
- Closed-loop control ensures accuracy and stability even under varying loads.
- Readily available with good documentation and integration support.

Conclusion: The NEMA 24 closed-loop stepper motor with a 10:1 planetary gearbox meets all key motion and performance requirements of the AMR platform, making it a robust and efficient choice for reliable autonomous operation.

6.5 Inertial Measurement Unit (IMU)

Selected: Bosch BNO055

Key Characteristics:

- Integrated 9-axis sensor: 3-axis accelerometer, 3-axis gyroscope, 3-axis magnetometer
- Onboard Cortex-M0+ microcontroller running sensor fusion algorithms
- Output data: fused orientation (quaternions, Euler angles), linear acceleration, gravity vector
- Interfaces: I²C, UART, HID-I²C; logic voltage 1.7 V to 3.6 V
- Accelerometer ranges: $\pm 2g$, $\pm 4g$, $\pm 8g$, $\pm 16g$
- Gyroscope ranges: $\pm 125^\circ\text{C/s}$ to $\pm 2000^\circ\text{C/s}$
- Magnetometer typical range: $\pm 1300\ \mu\text{T}$ (X,Y), $\pm 2500\ \mu\text{T}$ (Z)
- Operating voltage: 2.4 V to 3.6 V
- Power modes: Normal, Low Power, Suspend
- Operating temperature: -40C to +85C

Justification:

- Provides fully fused absolute orientation data, reducing processing load on main controller
- Compact and lightweight package suitable for mobile robotics
- Wide operating voltage and temperature range for robustness
- Multiple communication interfaces for flexible integration

- Automatic calibration and self-test features improve reliability

Alternatives considered: MPU-9250, LSM9DS1, ICM-20948

Price: Rs. 3,652.00



Figure 13: Inertial Measurement Unit

Provides accurate orientation data with onboard sensor fusion.

6.6 Communication Module

Selected: Mercusys MW300UH 300Mbps High Gain Wireless USB Adapter

Key Characteristics:

- **Antenna Type:** External high gain antenna
- **Dimensions:**
 - Without antennas: $113.2 \times 63.8 \times 21.7$ mm
 - With vertical antennas: $113.2 \times 100.5 \times 201.1$ mm
- **Signal Rate:**
 - 11n: Up to 300 Mbps (dynamic)
 - 11g: Up to 54 Mbps (dynamic)
 - 11b: Up to 11 Mbps (dynamic)
- **Wireless Standards:** IEEE 802.11n/g/b
- **Frequency Range:** 2.400 - 2.4835 GHz
- **Reception Sensitivity:** Ranges from -68 dBm to -94 dBm depending on data rate
- **Interfaces:** Micro USB 2.0

- **Transmission Power:** ≤ 20 dBm (EIRP)
- **Security:** WEP, WPA/WPA2, WPA-PSK/WPA2-PSK
- **Operating Environment:**
 - Temperature: 0C to 40C operating, -40C to 70C storage
 - Humidity: 10% to 90% non-condensing (operating)

Justification:

- Offers reliable wireless communication, critical for remote monitoring and diagnostics.
- Supports high-speed wireless communication up to 300 Mbps suitable for data transfer needs
- External antenna provides extended range and improved signal quality
- Compatible with common operating systems ensuring easy integration
- Compact form factor with standard Micro USB interface simplifies hardware integration

Alternatives considered: TP-Link TL-WN725N, Netgear Nighthawk A7000

Price: Rs. 4,700.00



Figure 14: Communication module

6.7 User Interface

Selected: 7-Inch IPS HDMI-Compatible Touch Screen Display

Key Characteristics:

- **Display Type:** 7-inch IPS LCD with capacitive touch

- **Resolution:** 1024×600 pixels
- **Touch Technology:** Capacitive multi-touch
- **Interface:** HDMI input with included HDMI cable
- **Viewing Angles:** Wide viewing angles due to IPS technology
- **System Compatibility:** Supports Raspbian, Ubuntu, Windows, and other OS
- **Package Contents:** Display, HDMI cable, mounting nuts

Justification:

- Provides high-quality visuals with sharp resolution and vibrant colors
- Capacitive touch allows smooth and responsive user interaction
- HDMI compatibility ensures easy connection with Raspberry Pi and other devices
- Compact size fits well for portable and embedded applications in the robot

Price: Rs. 10,890.00



Figure 15: HDMI Touch Screen Display

6.8 Mechanical Components

Driving Wheels: 2 units

- **Type:** Rubber treaded wheels for good traction
- **Diameter:** 20 cm
- **Material:** Rubber with durable aluminium hub
- **Mounting:** Directly connected to motor shafts on each side, positioned at the middle of the chassis
- **Load Capacity:** Suitable for carrying robot payload and weight

- **Justification:**

- Central placement allows balanced drive force and stable movement
- Direct drive reduces mechanical losses and improves responsiveness
- Rubber tread ensures good grip on indoor surfaces for precise navigation

Castor Wheels: 4 units

- **Type:** Swivel castor wheels for multi-directional support
- **Diameter:** 5 cm
- **Material:** Hard plastic or polyurethane for smooth rolling
- **Mounting:** Fixed at four chassis corners
- **Load Capacity:** Designed to stabilize the chassis and support overall weight
- **Justification:**
 - Corner placement ensures maximum stability and balance
 - Swivel design facilitates easy turning and maneuvering
 - Reduces friction during directional changes, preserving floor surfaces



Figure 16: Driving Wheel



Figure 17: Caster Wheel

7 Power Management Modules

To ensure stable and efficient power delivery to various subsystems within the AMR robot, a set of DC-DC power converters have been incorporated. These modules are responsible for stepping up or stepping down voltages to match the requirements of individual components such as microcontrollers, sensors, computing units, and motor drivers. All selected modules are compact, high-efficiency, and designed to operate reliably under variable load conditions. Two of them feature waterproof enclosures, making them suitable for rugged environments and improving the robot's resilience to external factors like dust and humidity.

24V to 12V Step-Down Converter Module

Step-down (buck) converter designed to convert 24 V DC input to stable 12 V DC output with adjustable current up to 5 A. Features a design suitable for rugged environments and reliable power regulation. This is used to supply the power to the LIDAR.

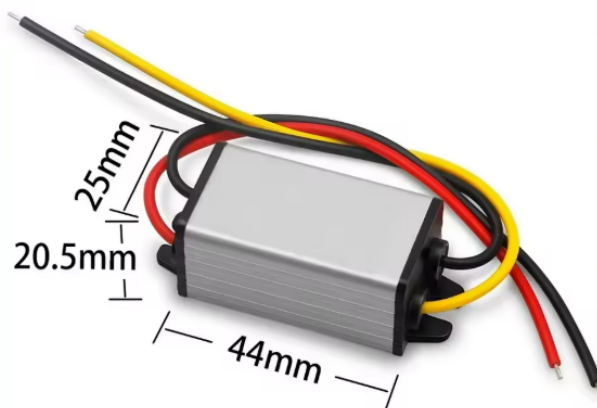


Figure 18: 24V to 12V Step-Down Converter Module

DC-DC Booster- Adjustable Step-Up Converter

High-power boost converter module featuring a wide input range of 10 V to 60 V DC and an adjustable output from 12 V up to 90 V DC. Delivers up to 20 A output current (30 A max input current from 10–30 V or 25 A from 31–60 V). Compact and robust, ideal for high-power applications needing flexible, step-up power regulation. This is used to supply power to the motors.



Figure 19: DC-DC Booster- Adjustable Step-Up Converter

B560C 5A DC-DC Buck Converter Module

High-current buck converter module accepting 5 V to 60 V input voltage, providing adjustable output voltage from 5 V to 20 V with a maximum output current of 5 A. Features low ripple voltage (~ 3 mV), suitable for high-performance power regulation. This is used to supply power to the Jetson Nano.

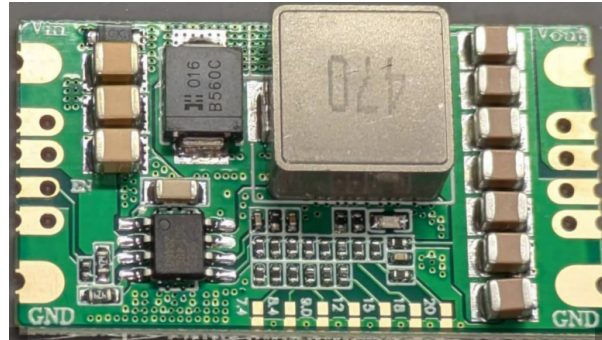


Figure 20: B560C 5A DC-DC Buck Converter Module

DC-DC Step-Down Power Supply Module (7-100V to 5/9/12/24V, 2A)

Wide input voltage range buck converter module accepting 7 V to 100 V DC input, with selectable output voltages of 5 V, 9 V, 12 V, or 24 V DC and maximum output current of 2 A. Suitable for applications requiring wide input voltage adaptability. This is used to supply power to the MCU.

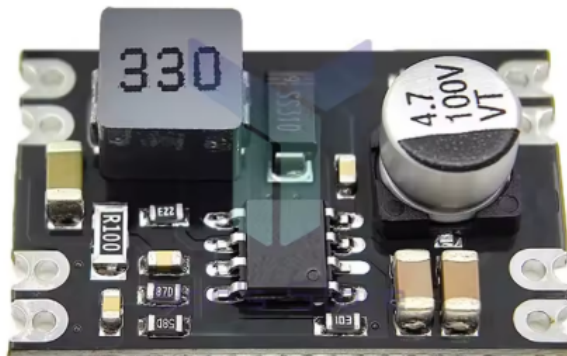


Figure 21: DC-DC Step-Down Power Supply Module

8 Conceptual Design

8.1 Conceptual Designs(Initial Sketches)

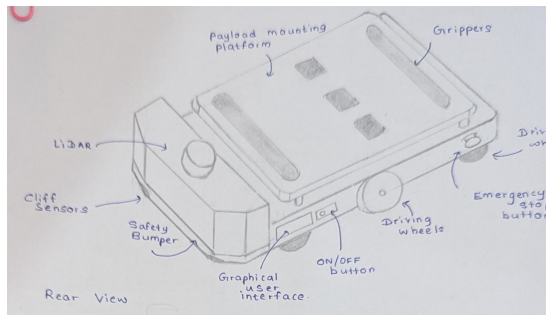


Figure 22: Initial Sketch 1

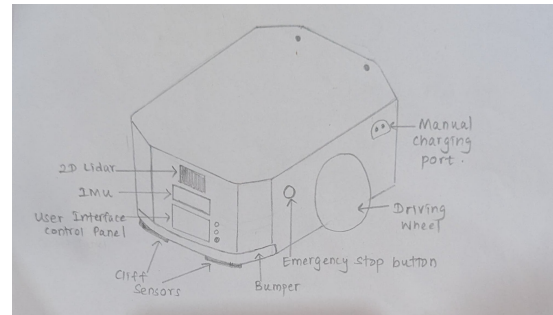


Figure 23: Initial Sketch 2

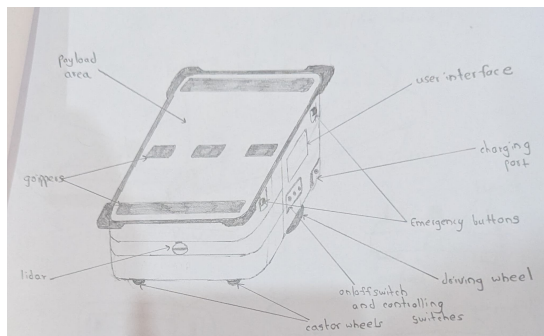


Figure 24: Initial Sketch 3

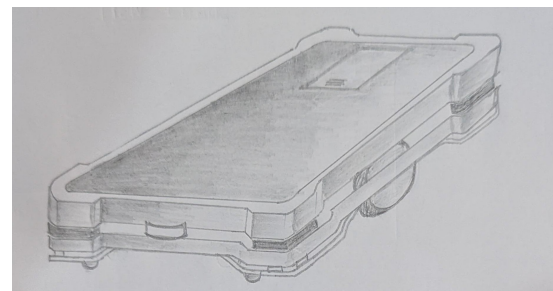


Figure 25: Initial Sketch 4

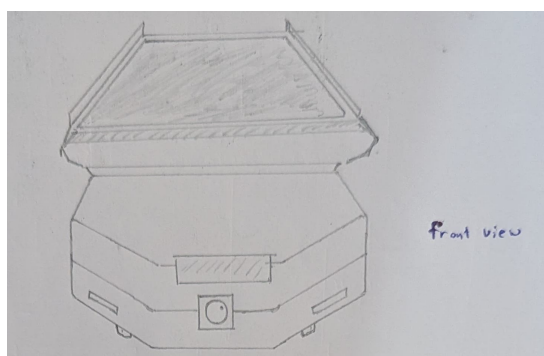


Figure 26: Initial Sketch 5

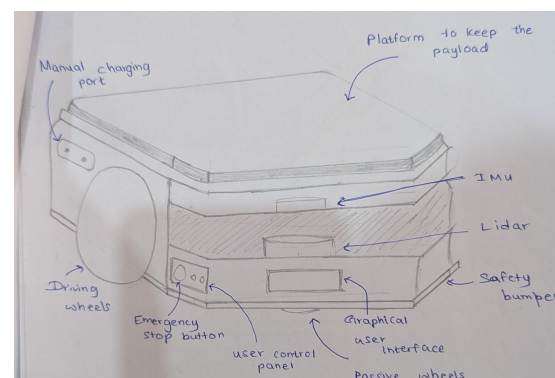


Figure 27: Initial Sketch 6

8.2 FINAL Conceptual Design

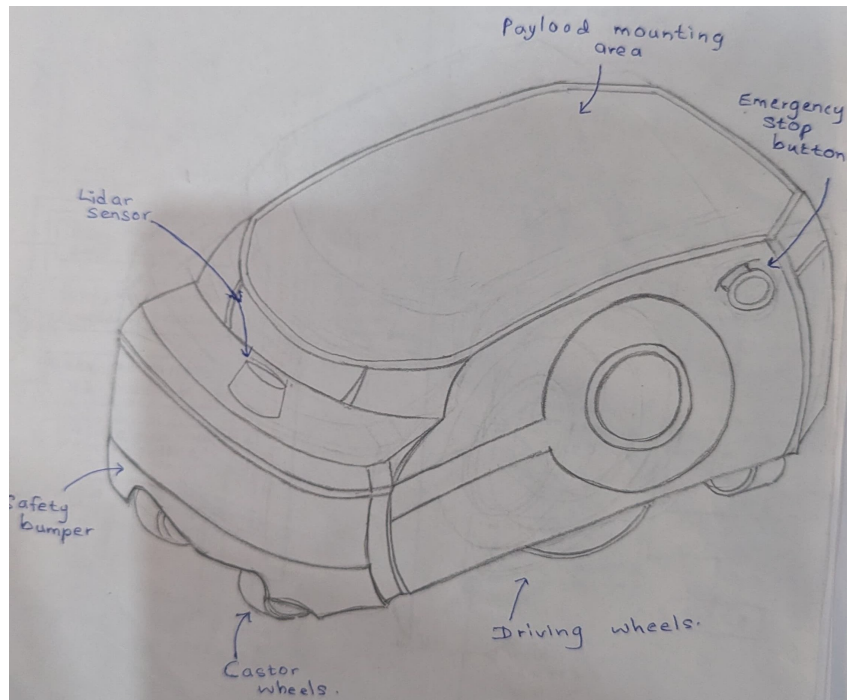


Figure 28: Front View

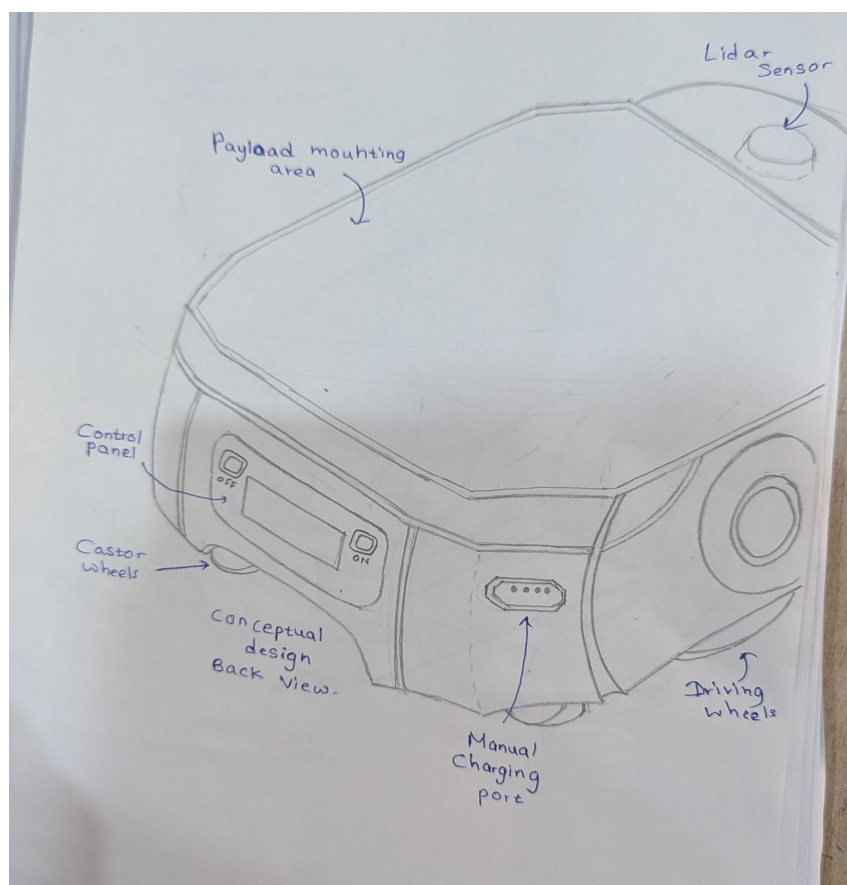


Figure 29: Rear View

9 Block Diagram

9.1 Initial Block Diagrams

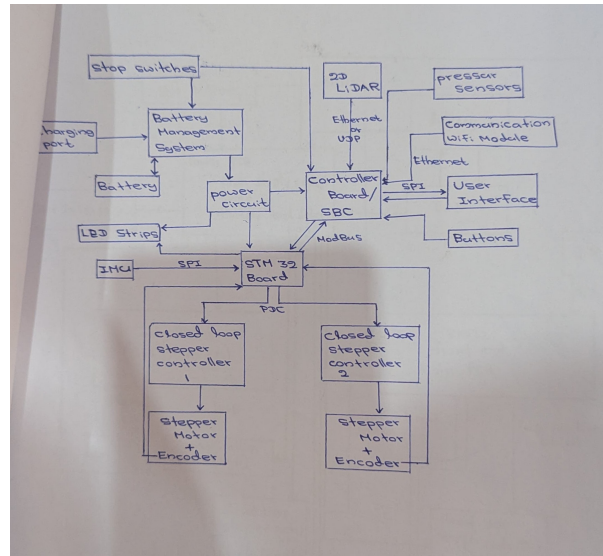


Figure 30: Iteration 1

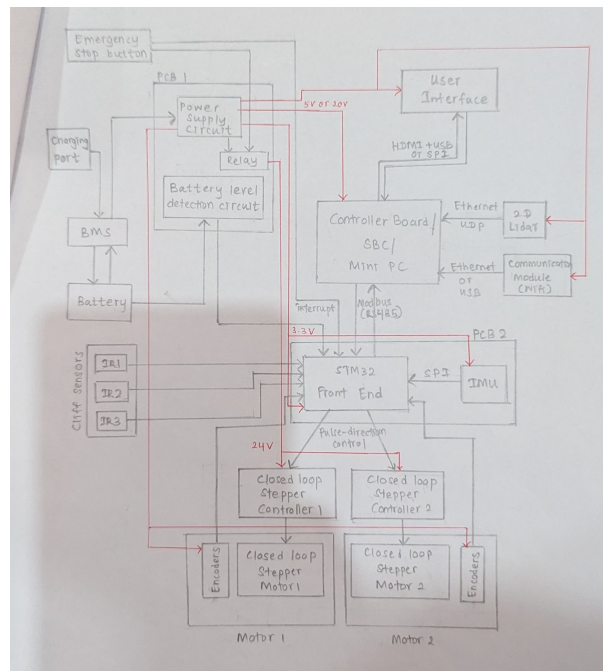


Figure 31: Iteration 2

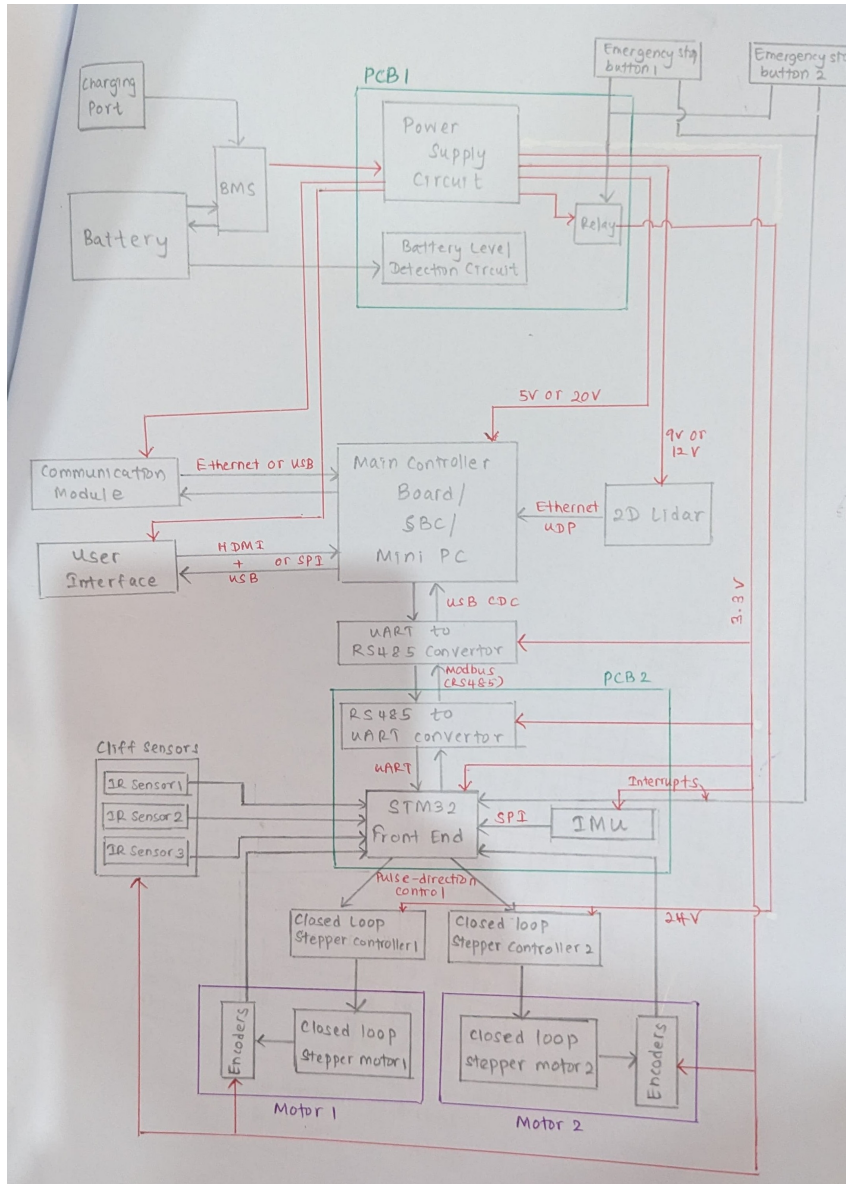


Figure 32: Iteration 3

9.2 Latest Block Diagram

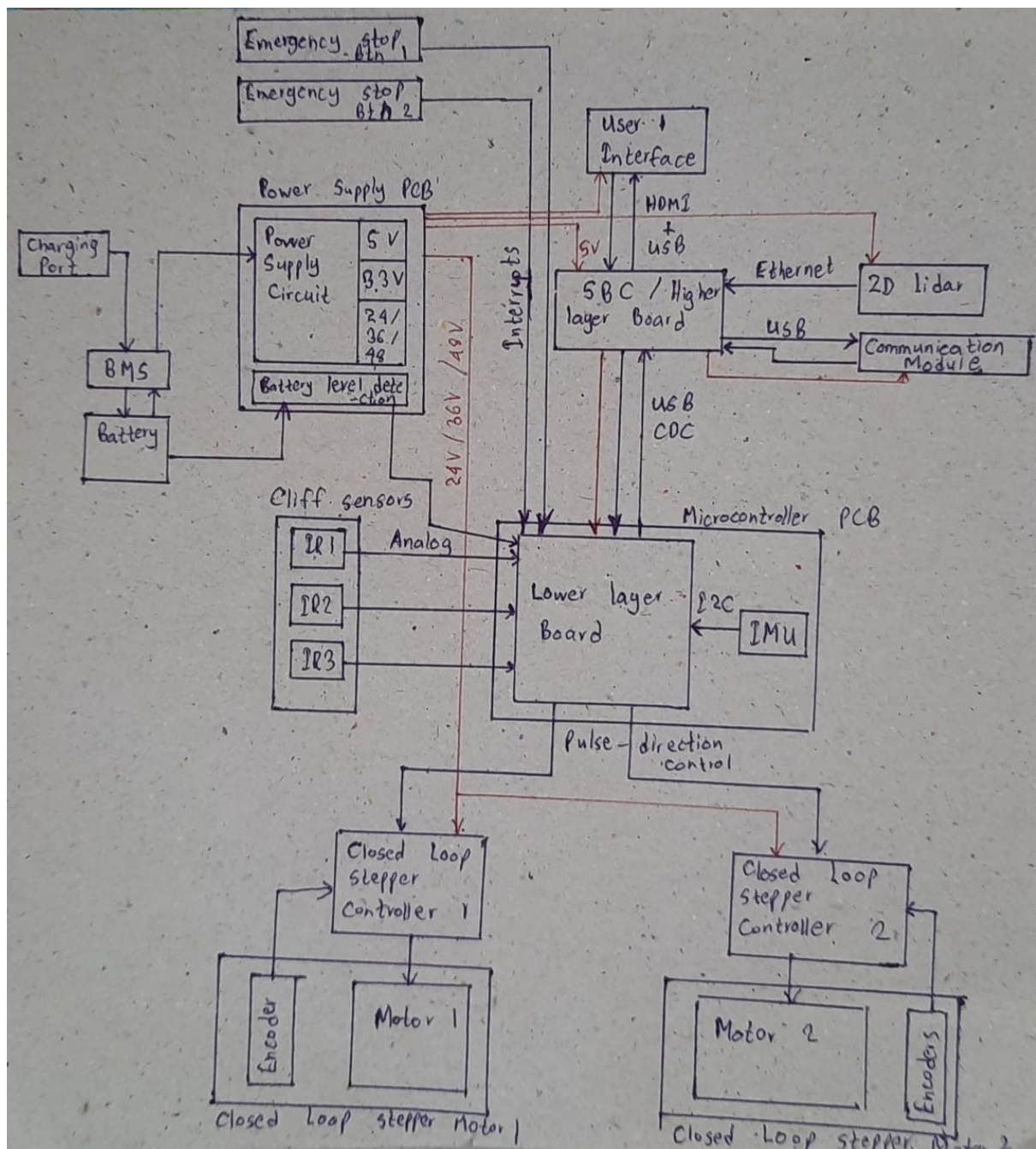


Figure 33: Latest Block Diagram

10 Enclosure Design

The enclosure design of our Autonomous Mobile Robot (AMR) has been developed with a strong focus on both functional utility and aesthetic appeal, creating a solution that is not only practical and durable but also visually aligned with modern industrial and commercial expectations. The design achieves a seamless balance between performance and presentation, ensuring the robot can operate reliably in real-world environments while projecting a clean, professional image.

To ensure cost efficiency without compromising on strength and functionality, the structural framework of the robot is based on a lightweight skeleton made from aluminium bars. These aluminium extrusions form the core internal frame of the AMR and are chosen for their excellent strength-to-weight ratio, corrosion resistance, and ease of machining. Aluminium also offers versatility in design and allows for modular construction, enabling easier customization or expansion in future iterations. The frame provides sufficient rigidity to withstand dynamic loads, vibrations, and minor impacts encountered during typical navigation tasks.

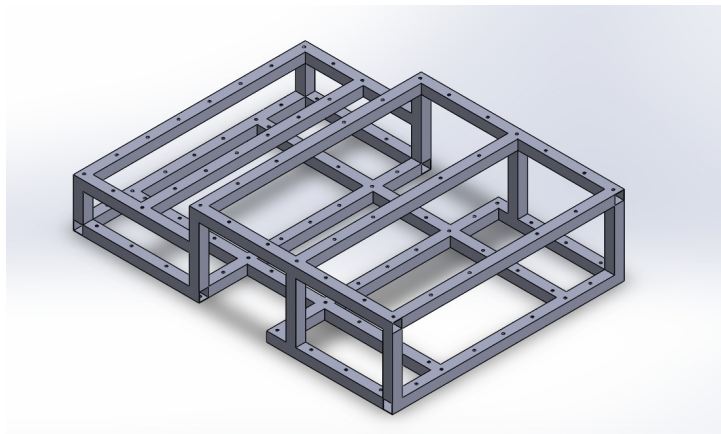


Figure 34: Metal Skeleton

The chassis is made from white steel. The use of steel in the chassis ensures structural stability and durability, capable of withstanding external forces and protecting sensitive electronics housed within the robot.

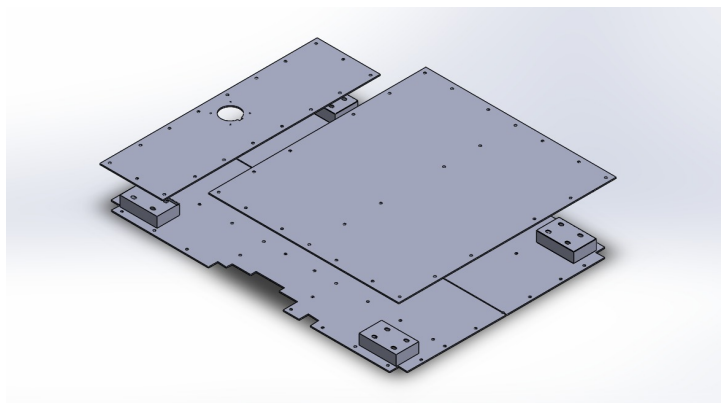


Figure 35: Steel Chassis

A defining feature of the enclosure is its external cover, also fabricated from white-coated steel, which serves several critical purposes. In addition to completing the robot's enclosed form and contributing to its professional aesthetic, this cover functions as the primary payload area. A dedicated mounting section is integrated into the top surface of the cover, enabling secure attachment of payloads or transport containers. The mounting platform is reinforced to handle variable loads while maintaining balance and stability, making the AMR adaptable to a wide range of use cases including material transport, delivery tasks, and modular tool systems. The removable top cover allows technicians to reach internal systems quickly without disassembling major structural elements.

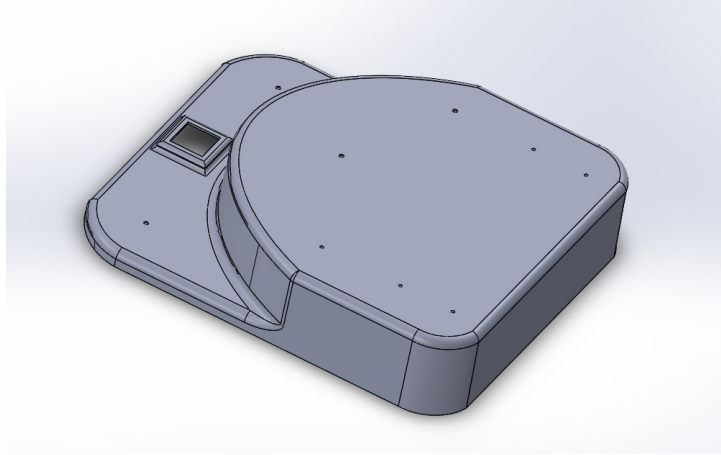


Figure 36: Steel External Cover

Our AMR robot features a differential drive system with two driving wheels positioned centrally on either side of the chassis. This central placement ensures balanced weight distribution and enables smooth rotation around the robot's axis, making it ideal for precise turning and maneuverability in narrow spaces. To support stability and maintain consistent ground contact, four caster wheels are placed at the front and rear corners. These passive wheels reduce tipping and help evenly distribute the robot's weight across the platform. This combination provides a stable yet agile design, ensuring both efficient mobility and the ability to handle varying floor conditions in indoor environments.

From a visual standpoint, the combination of polished aluminium edges, smooth white steel surfaces, and a clean geometric form factor gives the robot a sleek, high-tech appearance. This makes it suitable not only for industrial applications but also for customer-facing environments where design plays a role in user experience. The enclosure reflects attention to detail and engineering discipline, communicating both technological capability and aesthetic sensitivity.

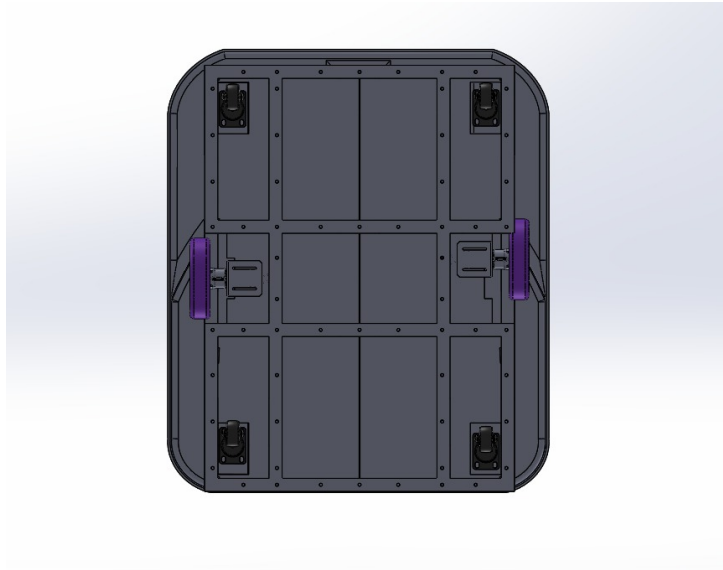


Figure 37: Wheel Placement

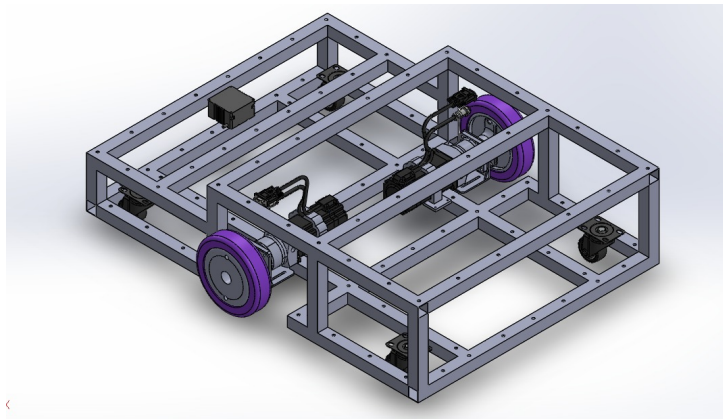


Figure 38: Placement of Motors and Wheels

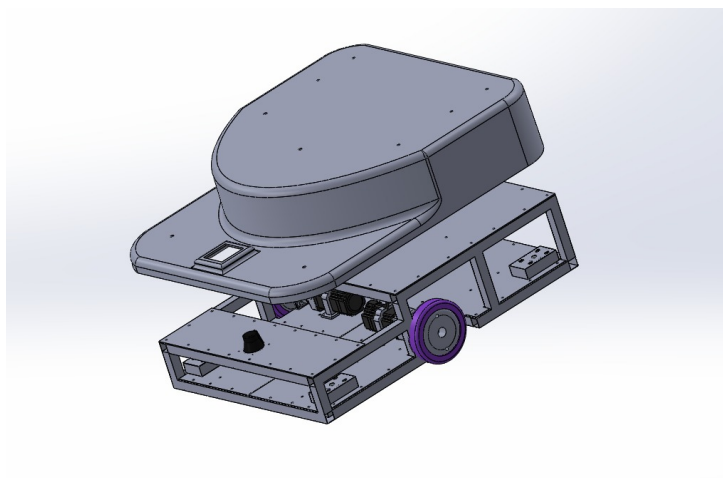


Figure 39: Side View

11 Schematic Design

11.1 Main Micro-controller Unit

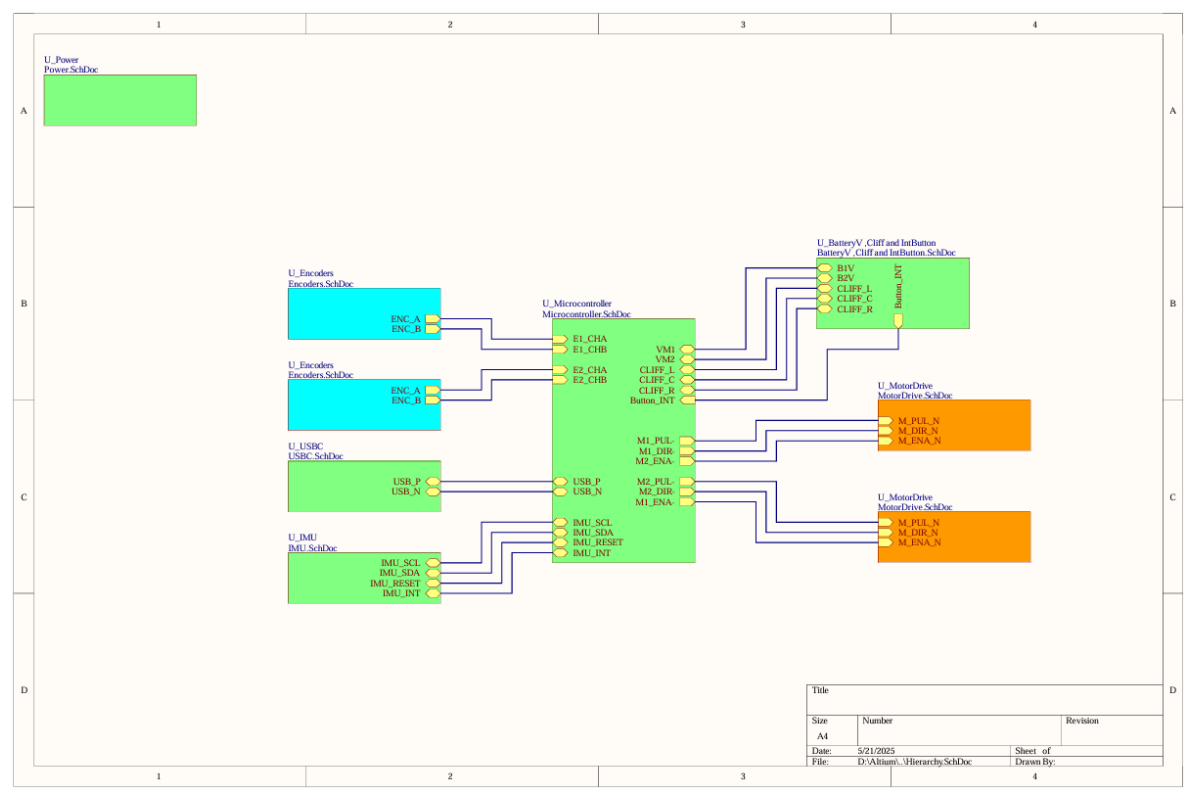


Figure 40: Hierarchical Design

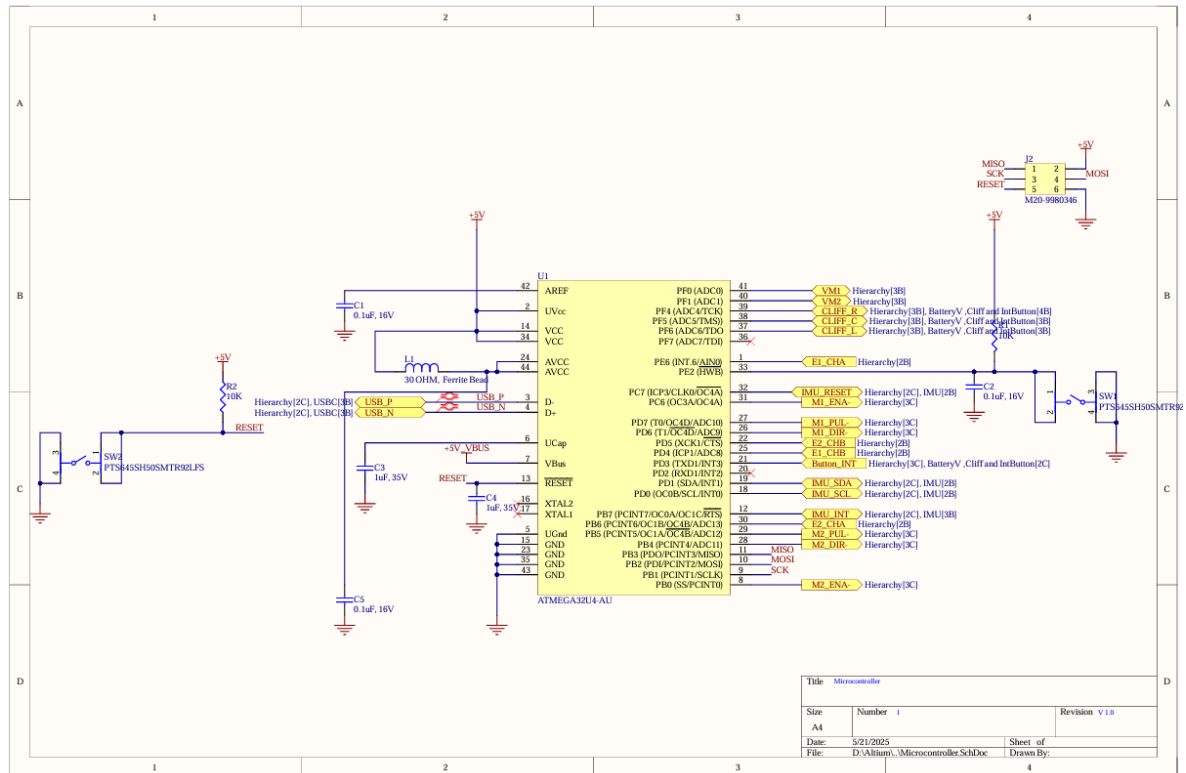


Figure 41: Micro-controller

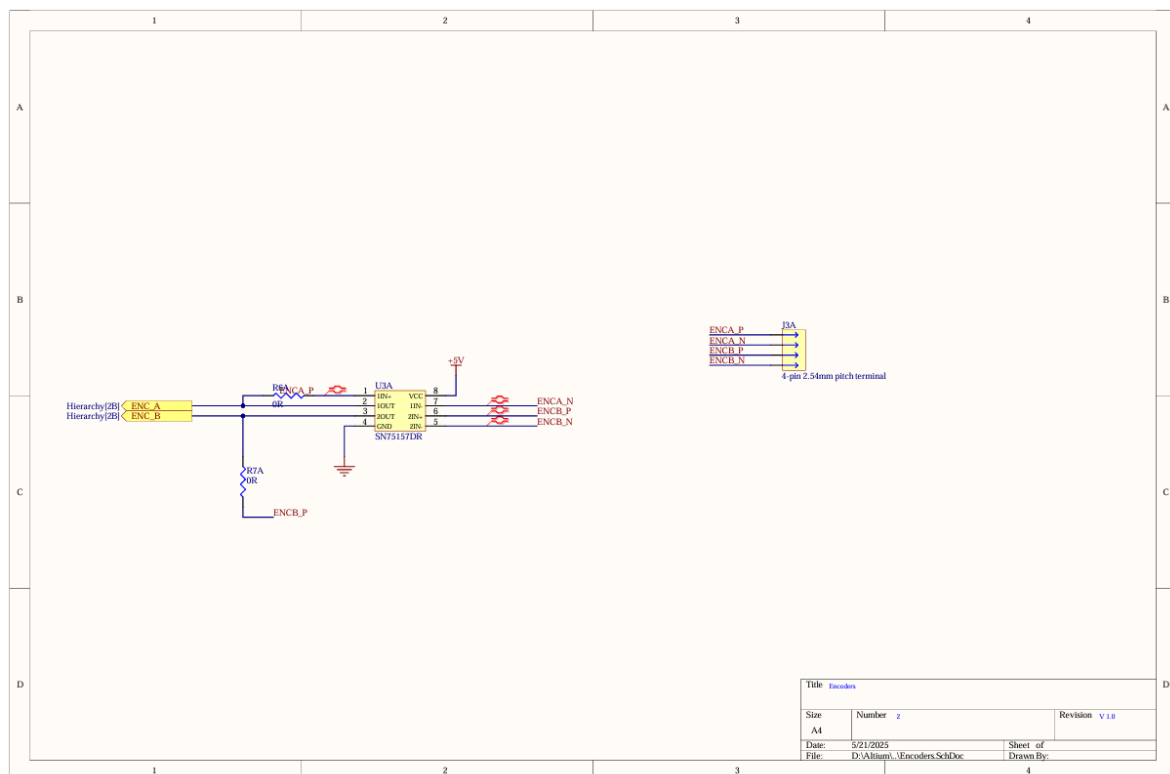


Figure 42: Encoder 1

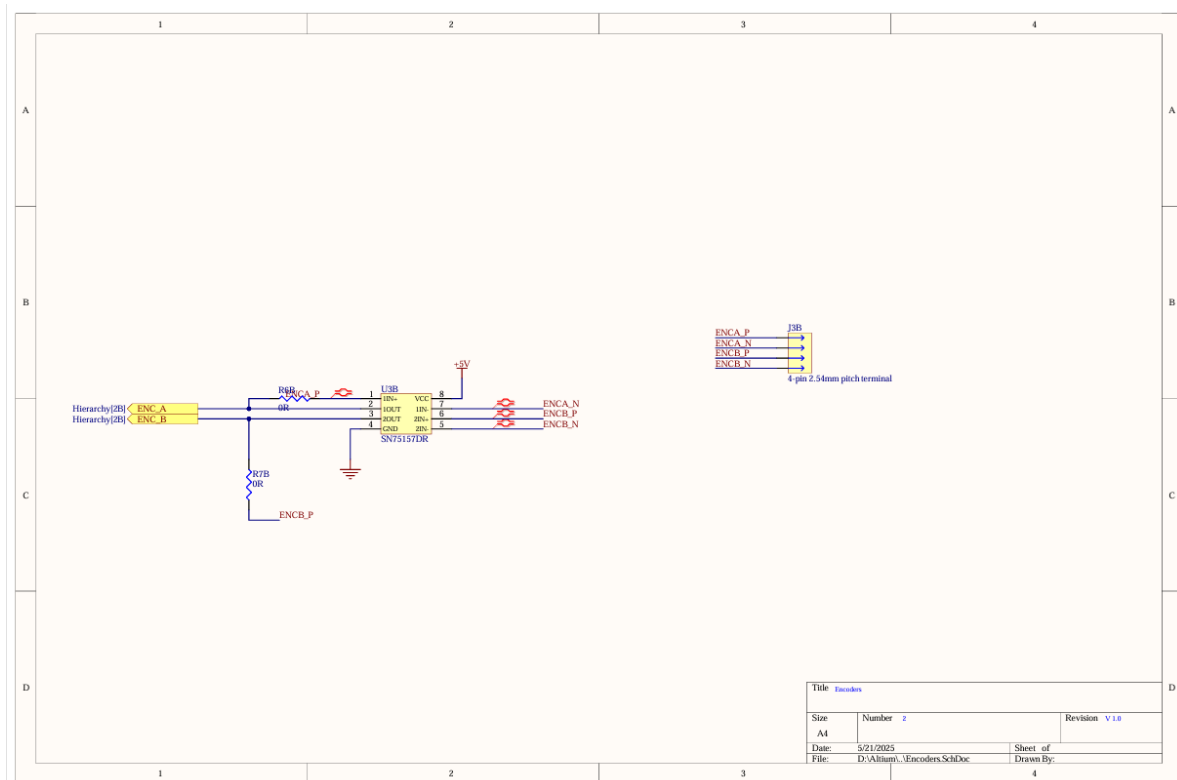


Figure 43: Encoder 2

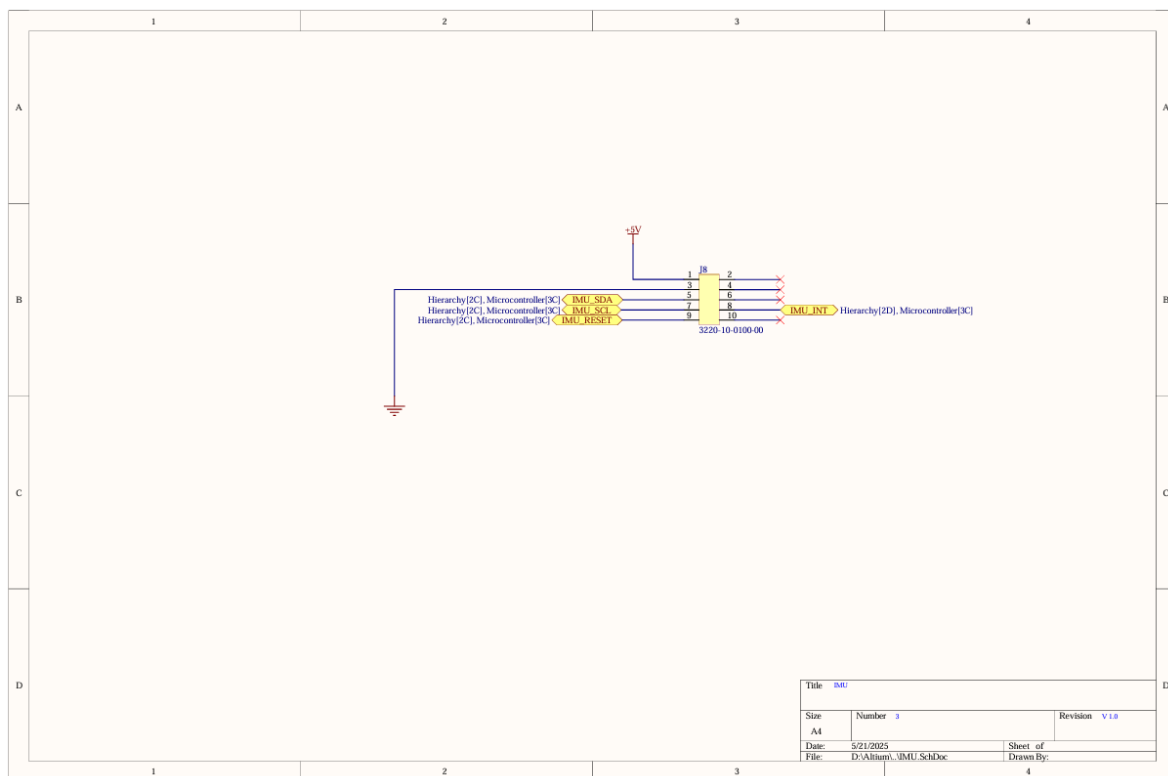


Figure 44: IMU

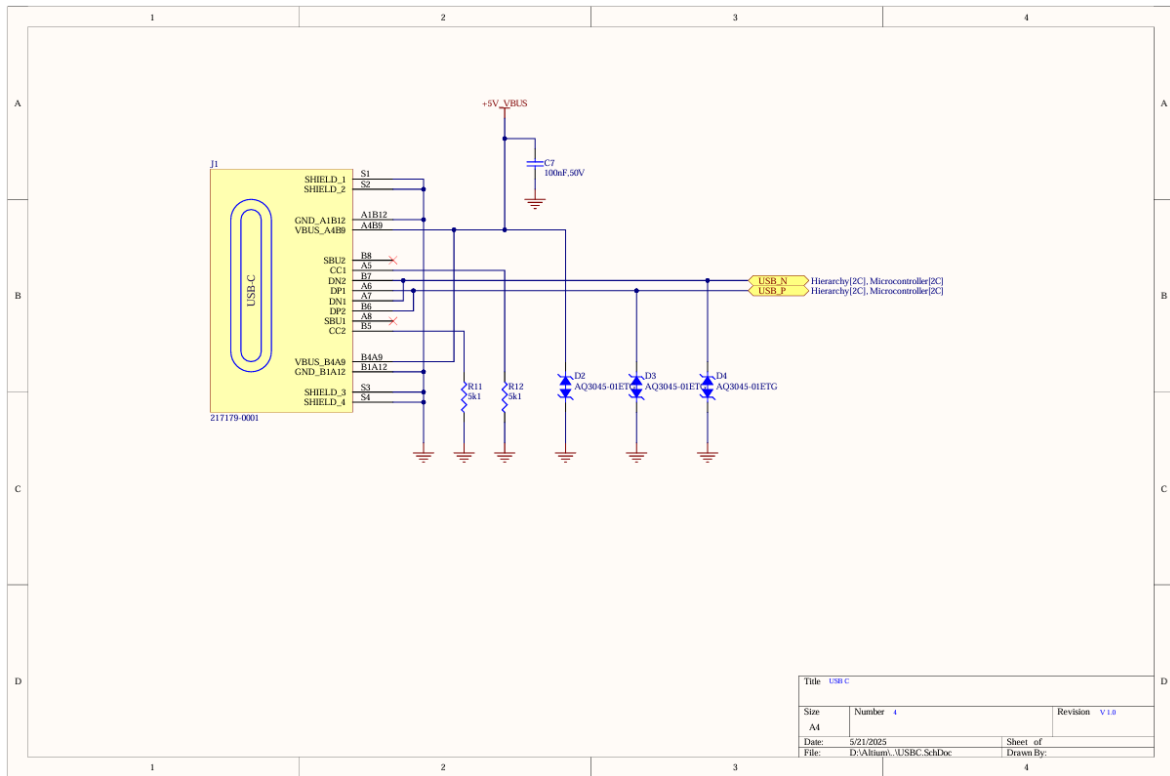


Figure 45: USB-C

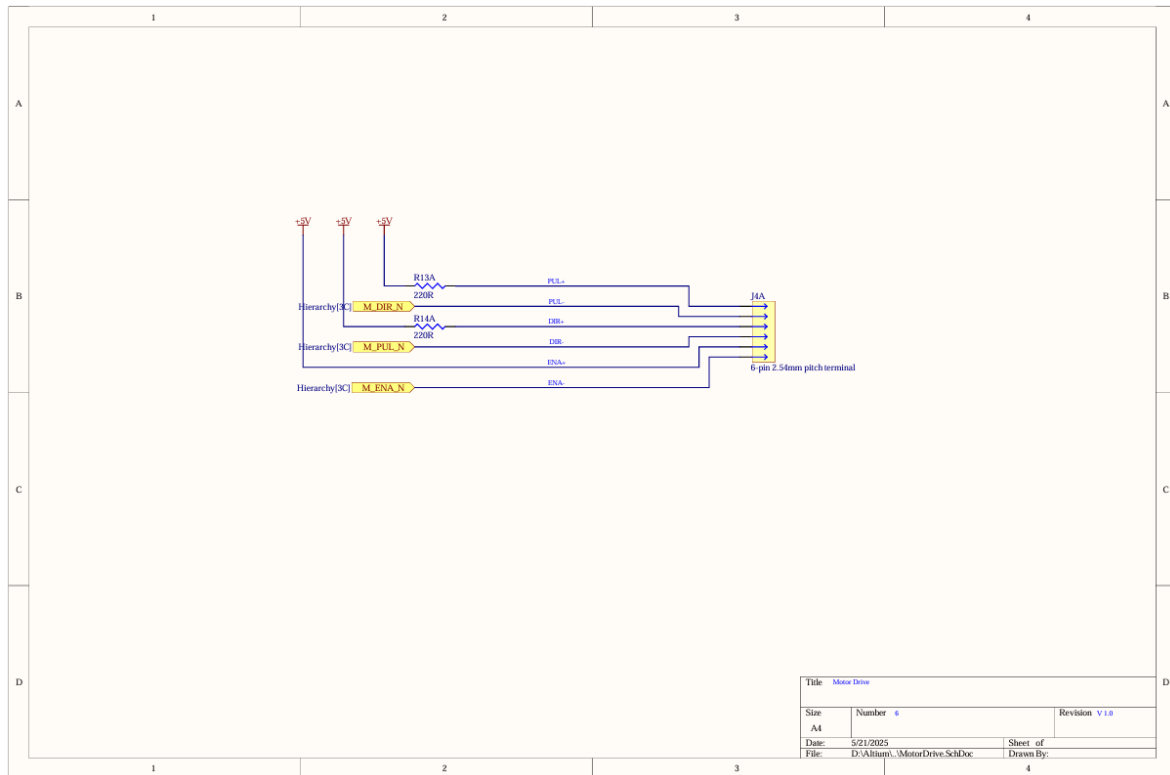
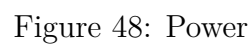


Figure 46: Motor Driver 1



All power for the control electronics and drive motors is derived from a single 24 V battery input on the PCB. From this bus, a boost converter raises the voltage to 36–40 V and distributes it to each motor driver. Concurrently, a dedicated buck regulator steps the 24 V down to 5 V to power the Jetson Nano, and that same 5 V rail is further tapped off and regulated down for the microcontroller unit. In a separate branch, a second buck converter drops the 24 V to 12 V to feed the LiDAR sensor.

11.3 Buffer Circuit

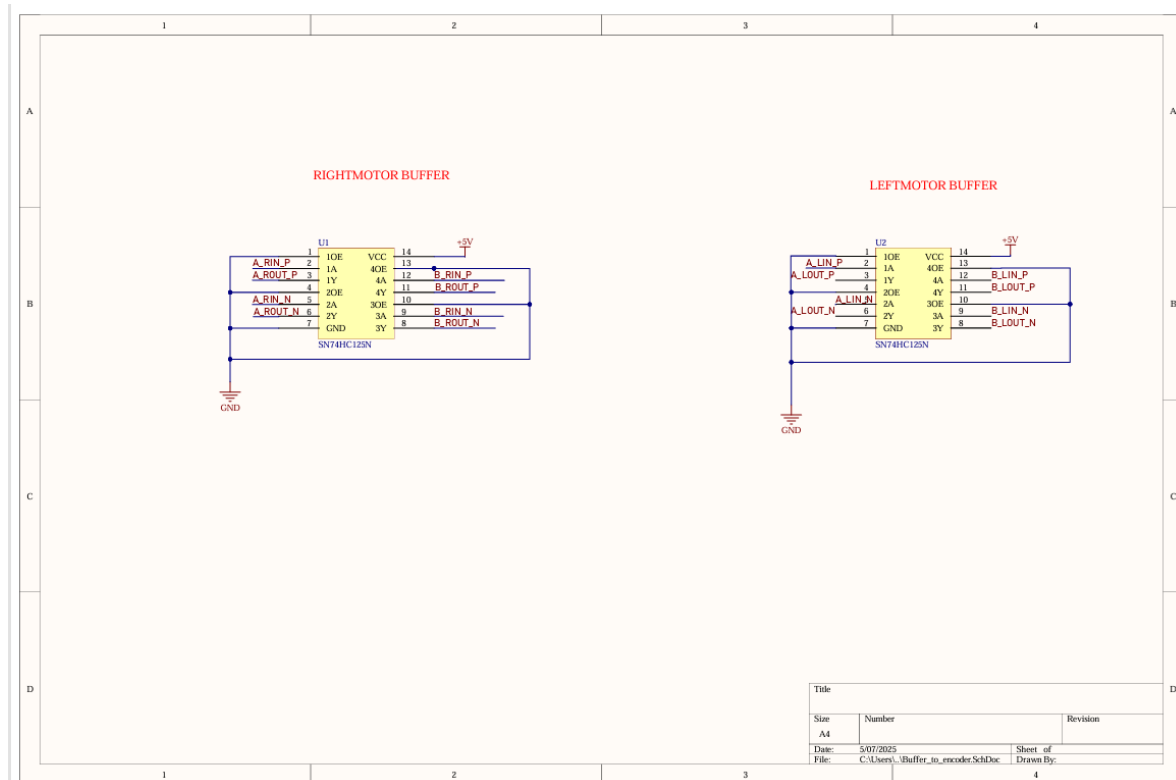


Figure 51: Buffer Circuit

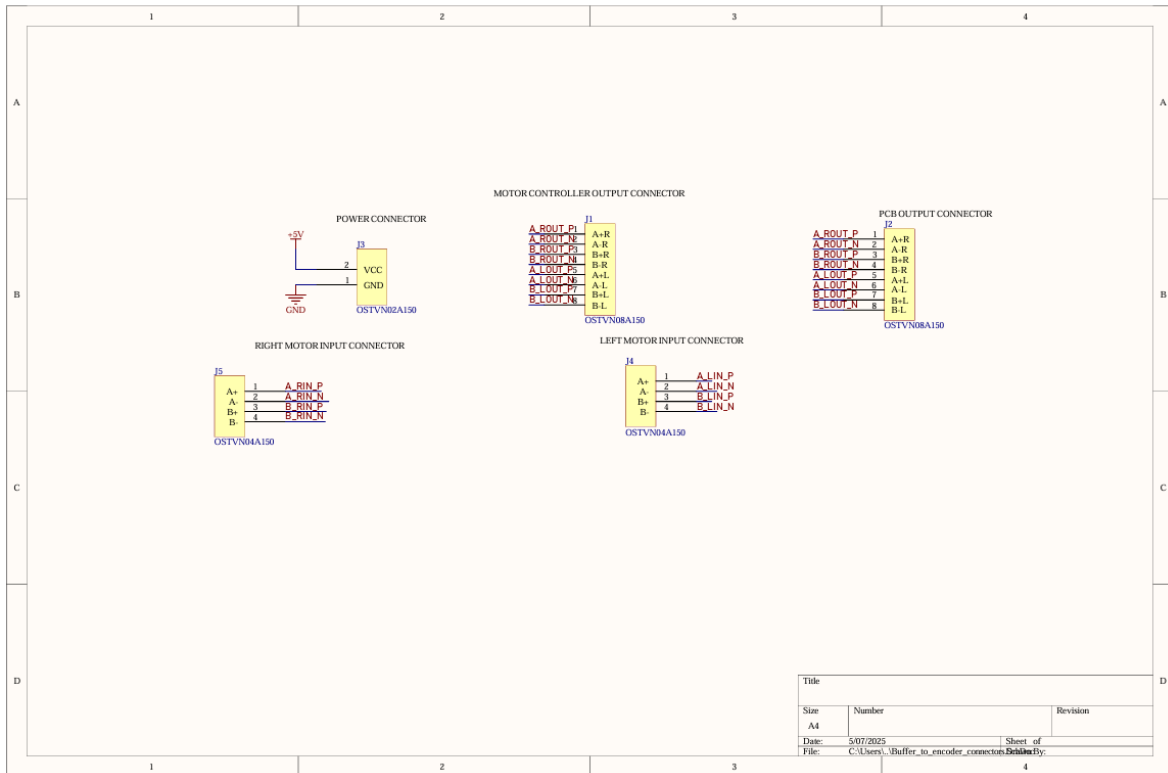


Figure 52: Buffer Circuit

11.4 IMU Circuit

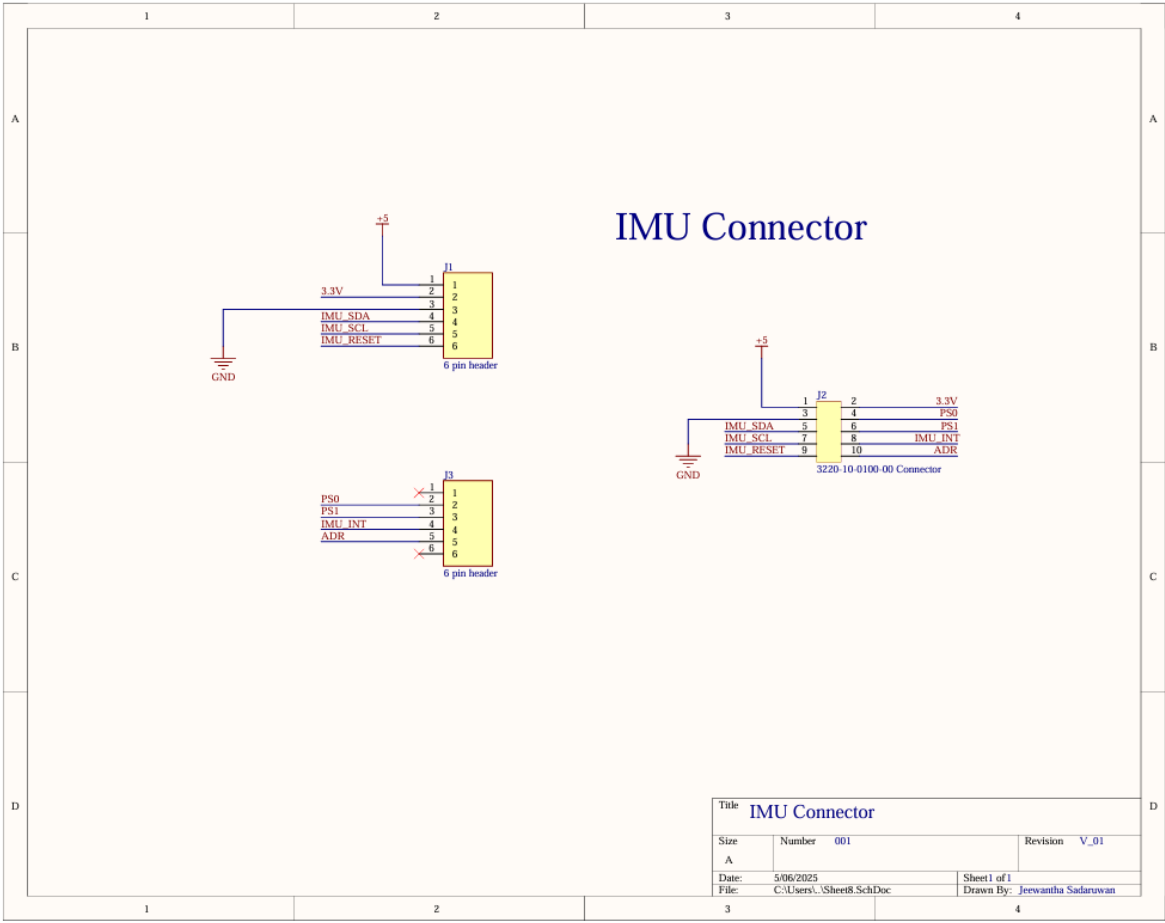


Figure 53: IMU Connector

12 PCB Design

12.1 Main PCB

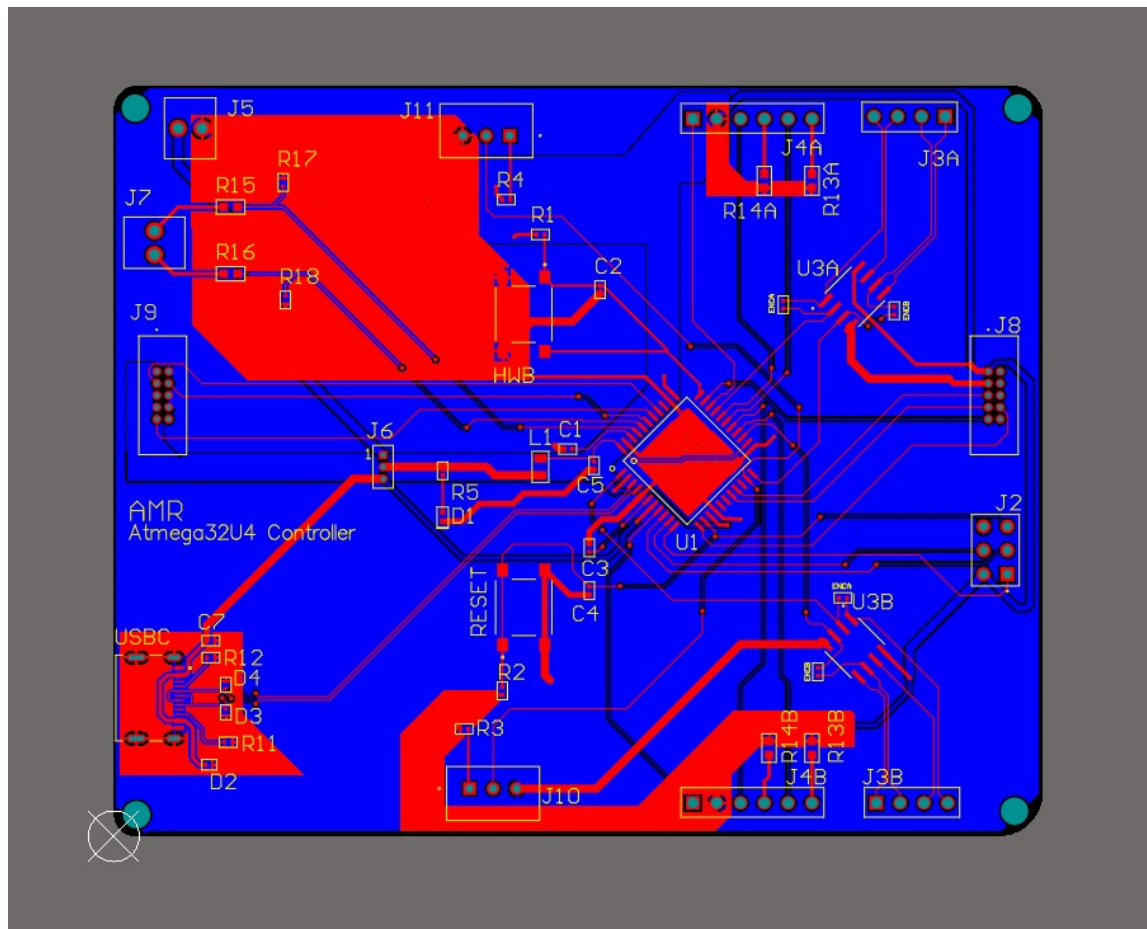


Figure 54: Main PCB Design

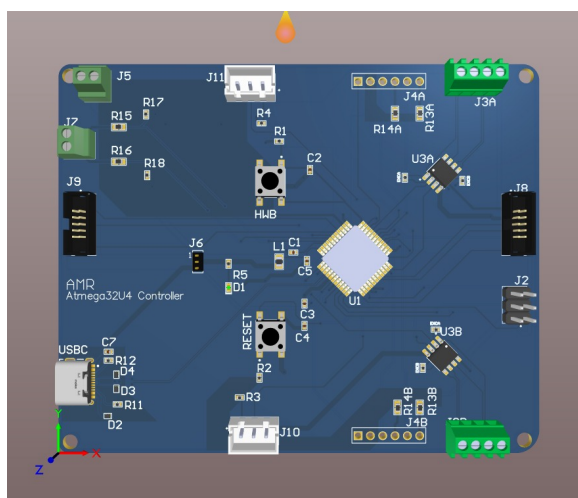


Figure 55: 3D Model Top View

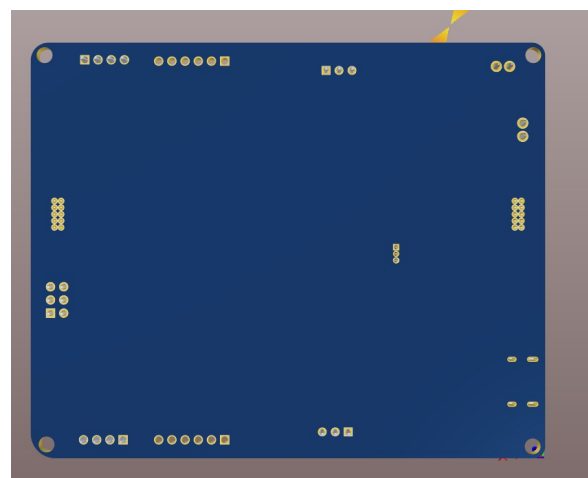


Figure 56: 3D Model Bottom View

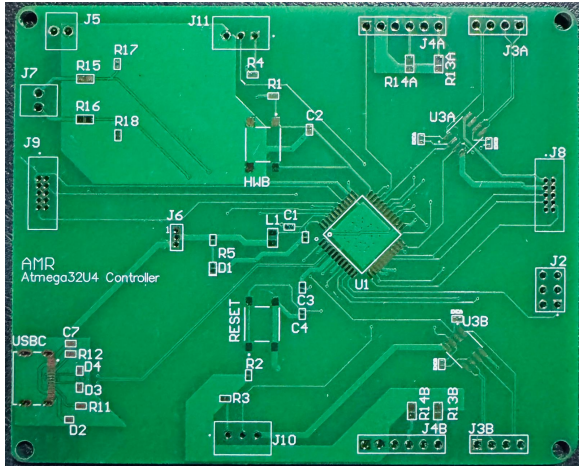


Figure 57: Bare PCB

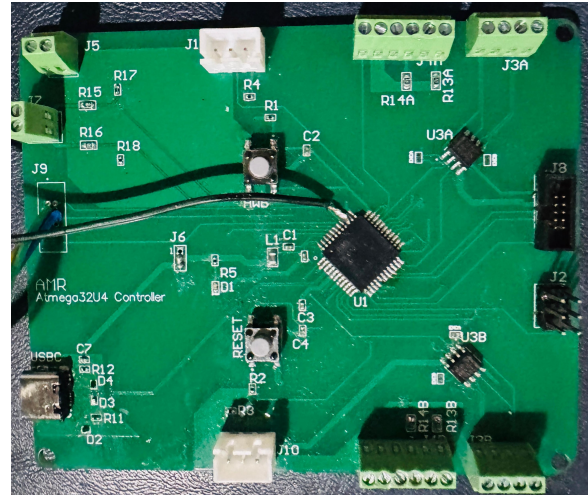


Figure 58: Soldered PCB

12.2 Power PCB

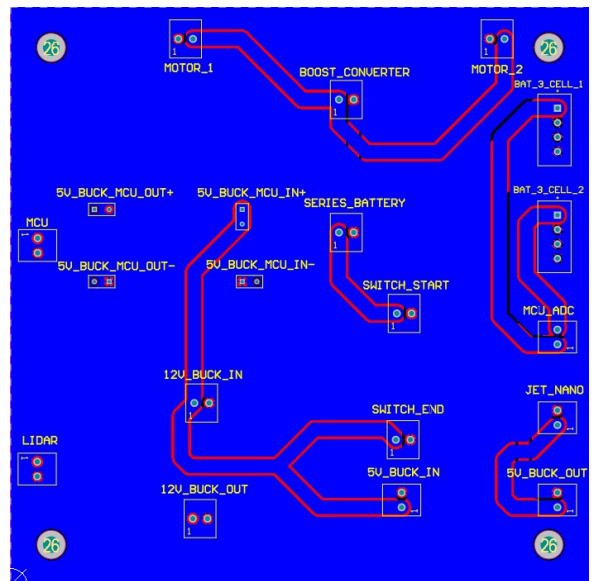
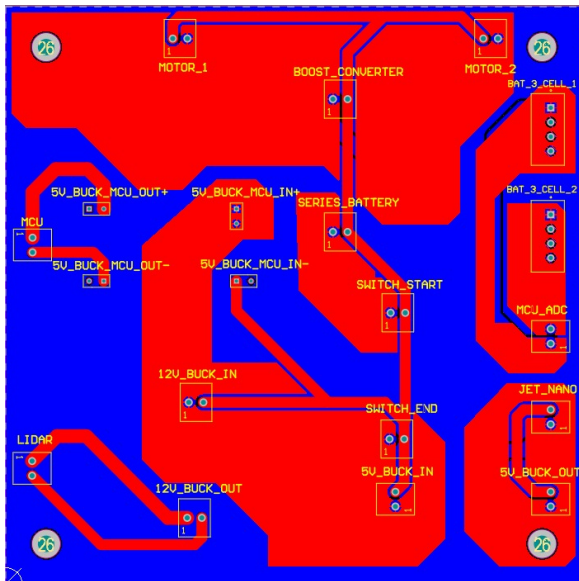


Figure 59: PCB Design

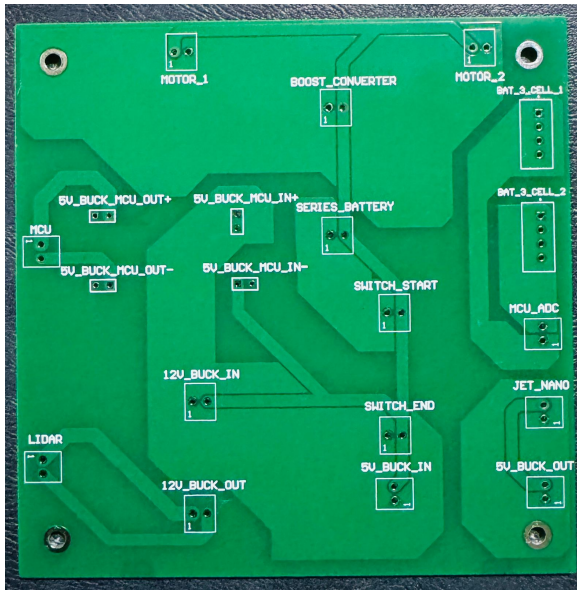


Figure 60: Bare PCB

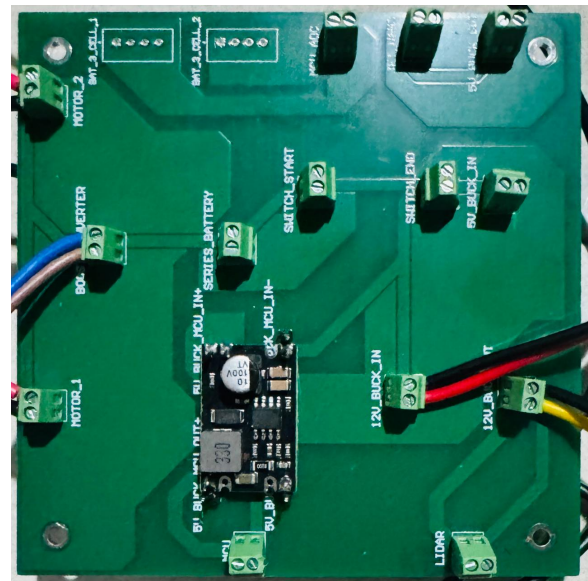


Figure 61: Soldered PCB

12.3 Buffer PCB

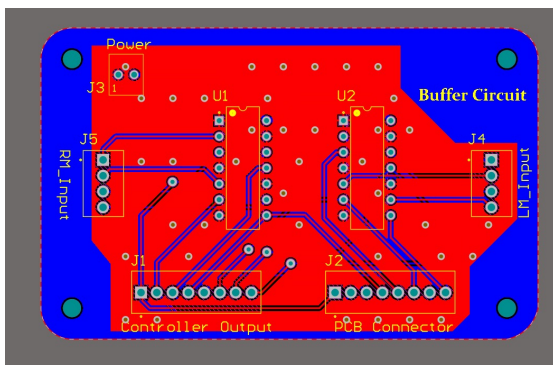


Figure 62: PCB Design

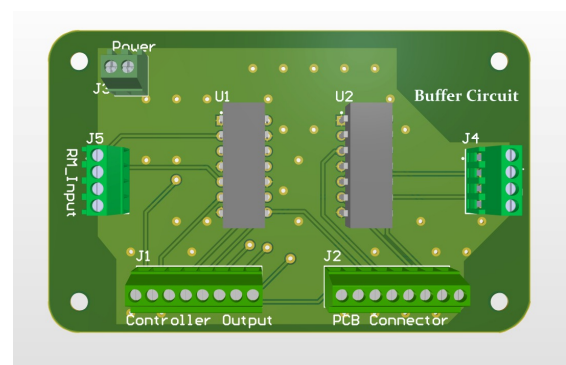


Figure 63: 3D Model

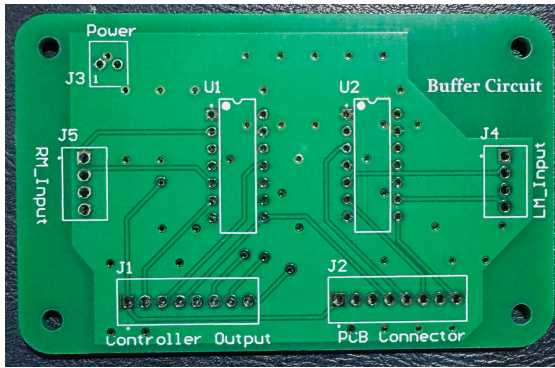


Figure 64: Bare PCB

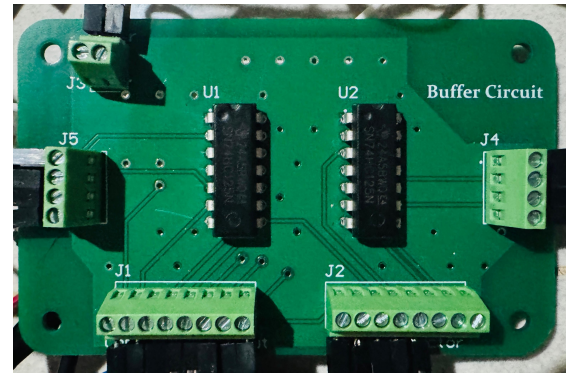


Figure 65: Soldered PCB

12.4 IMU Module PCB

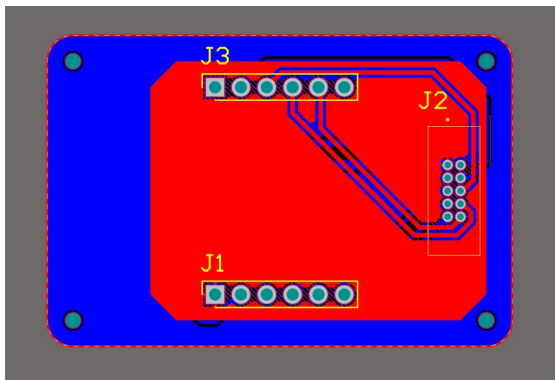


Figure 66: PCB Design

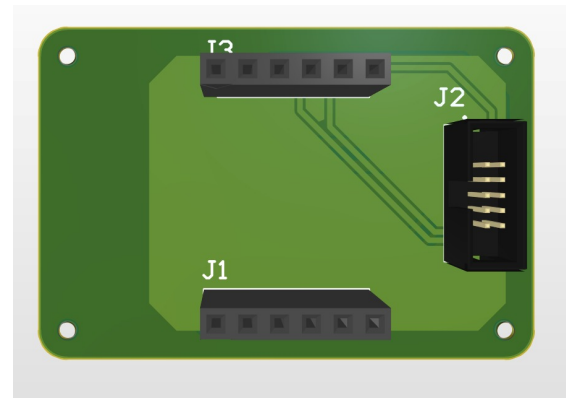


Figure 67: 3D Model

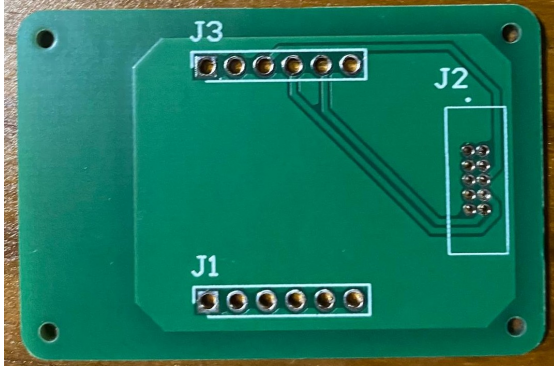


Figure 68: Bare PCB

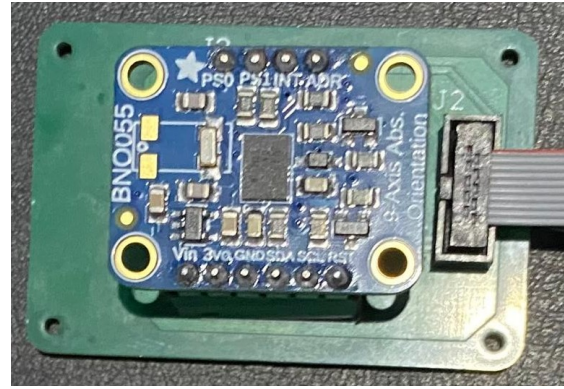


Figure 69: Soldered PCB

13 Safety Compliance

All operational zones around the robot are continuously monitored for hazards by integrated cliff sensors along the chassis front, which immediately halt motion if an unexpected drop is detected. In addition, large, easy-access emergency-stop buttons are mounted on both the sides, providing operators with an instant manual override to cut power to all drive and control systems. These redundant safety features ensure rapid cessation of movement in critical situations and compliance with ISO 13482 requirements for industrial mobile robots.

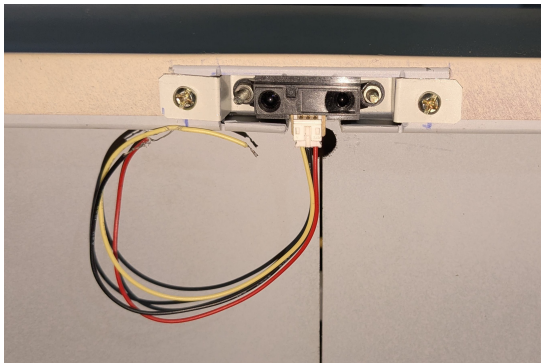


Figure 70: Cliff sensor placement



Figure 71: Emergency Stop Buttons

14 Construction of the Product

14.1 Mechanical Assembly and Enclosure

We began by constructing the skeleton of the AMR from aluminum bars, forming a rigid frame to support all subsequent components. The motor and gearbox mounting points were integrated into this frame with additional metal supports.

Next, zinc-aluminium coated metal sheets were attached to form the enclosure walls. Pillar filler was applied to seams and joints to ensure smooth surfaces prior to painting.



Figure 72: Skeleton frame constructed from aluminum bars.

14.2 Functional Component Placement

The two NEMA-24 stepper motors and planetary gearboxes were assembled and mounted via precision brackets, ensuring wheel alignment. A rigid coupler joins each motor shaft to its driving wheel hub.

Height-adjustable caster wheels at each corner maintain a level platform and allow fine-tuning of ground clearance.

Underneath the frame two cliff sensors were mounted on a dedicated cross-bar, providing reliable obstacle detection.

14.3 Integration of Electrical and Mechanical Systems

A SolidWorks-designed enclosure houses the power PCB and NVIDIA Jetson Nano. Adjacent to this, the battery compartment was fabricated and secured within the chassis. All wiring harnesses are routed through protective channels.



Figure 73: Structure after applying pillar



Figure 74: Structure after painting

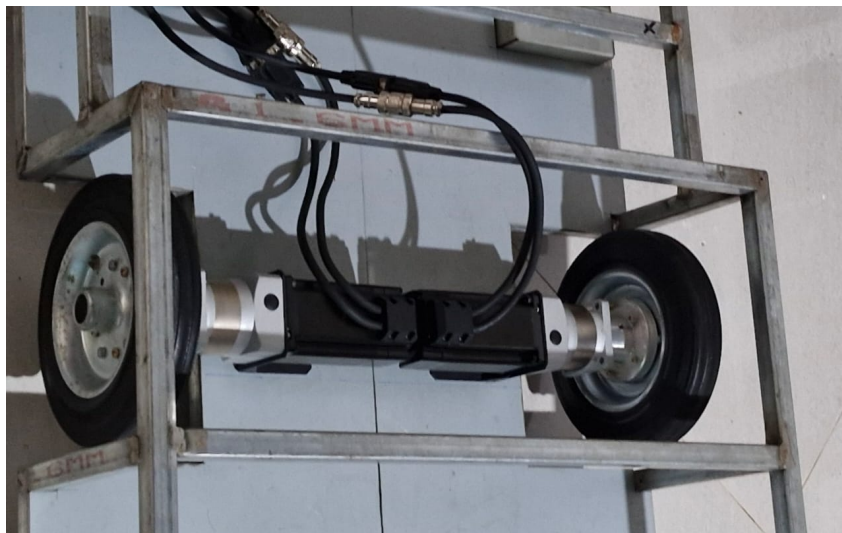


Figure 75: Motors, gearboxes and wheel placement.

Emergency stop buttons were integrated into the side panels via precision cutouts. A cutout for the touchscreen display ensures operator access.

14.4 Maintenance and Serviceability Features

All key components—motors, gearbox, electronics, and sensors—are accessible through removable panels. Quick-release fasteners are used at five main access points, minimizing downtime for inspections and repairs.



Figure 76: Driving wheel with shaft coupler.



Figure 77: Adjustable-height caster wheel

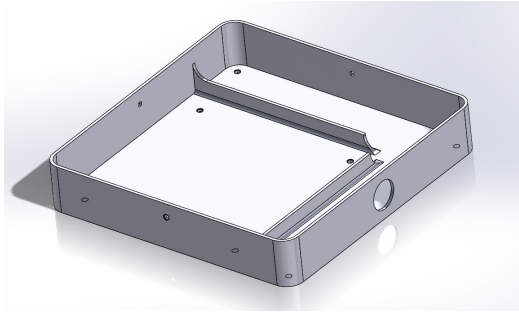


Figure 78: Power PCB Enclosure Bottom

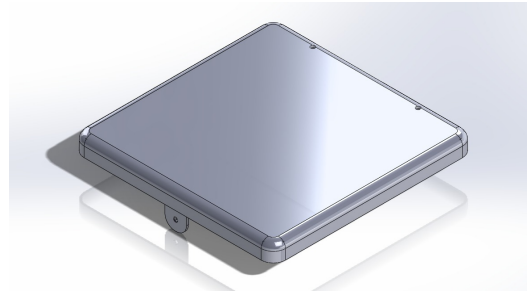


Figure 79: Power PCB Enclosure Lid

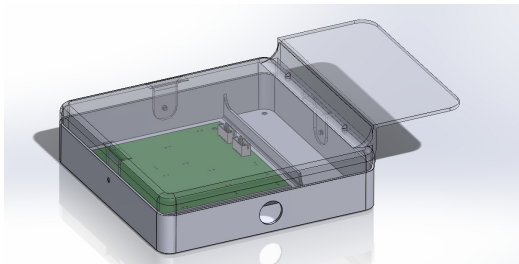


Figure 80: Power PCB Enclosure Assembly

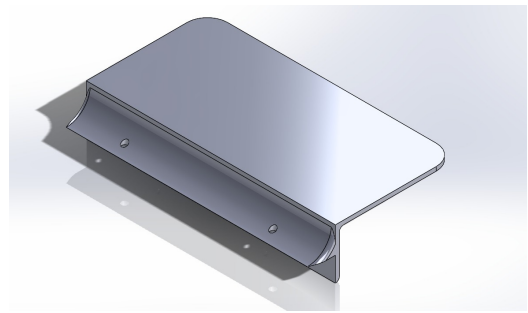


Figure 81: Battery Compartment

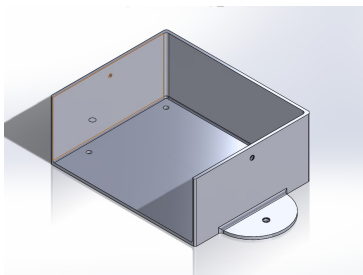


Figure 82: Jetson Nano Enclosure Bottom

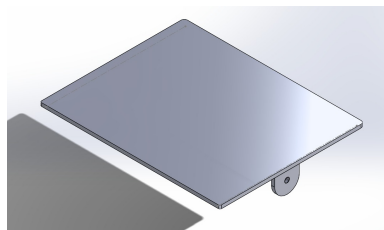


Figure 83: Jetson Nano Enclosure Top

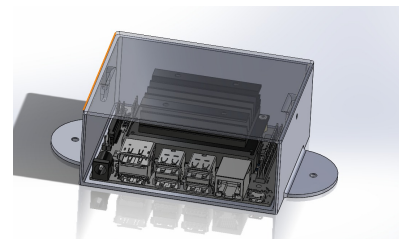


Figure 84: Jetson Nano Enclosure Assembly

15 Design Modifications After First Demonstration

Our Autonomous Mobile Robot (AMR) platform has undergone a series of targeted enhancements to optimize its performance, reliability, and usability. These updates encompass hardware refinements—such as upgraded PCB designs—as well as software improvements, enclosure design updates for more modular component layout, refined path-planning routines, and a more intuitive user-operator interface. Each of these developments has been carefully integrated and rigorously tested to ensure seamless operation in diverse operating environments. The details of these iterative improvements are given in the sections below.

15.1 Final System Hardware Block Diagram

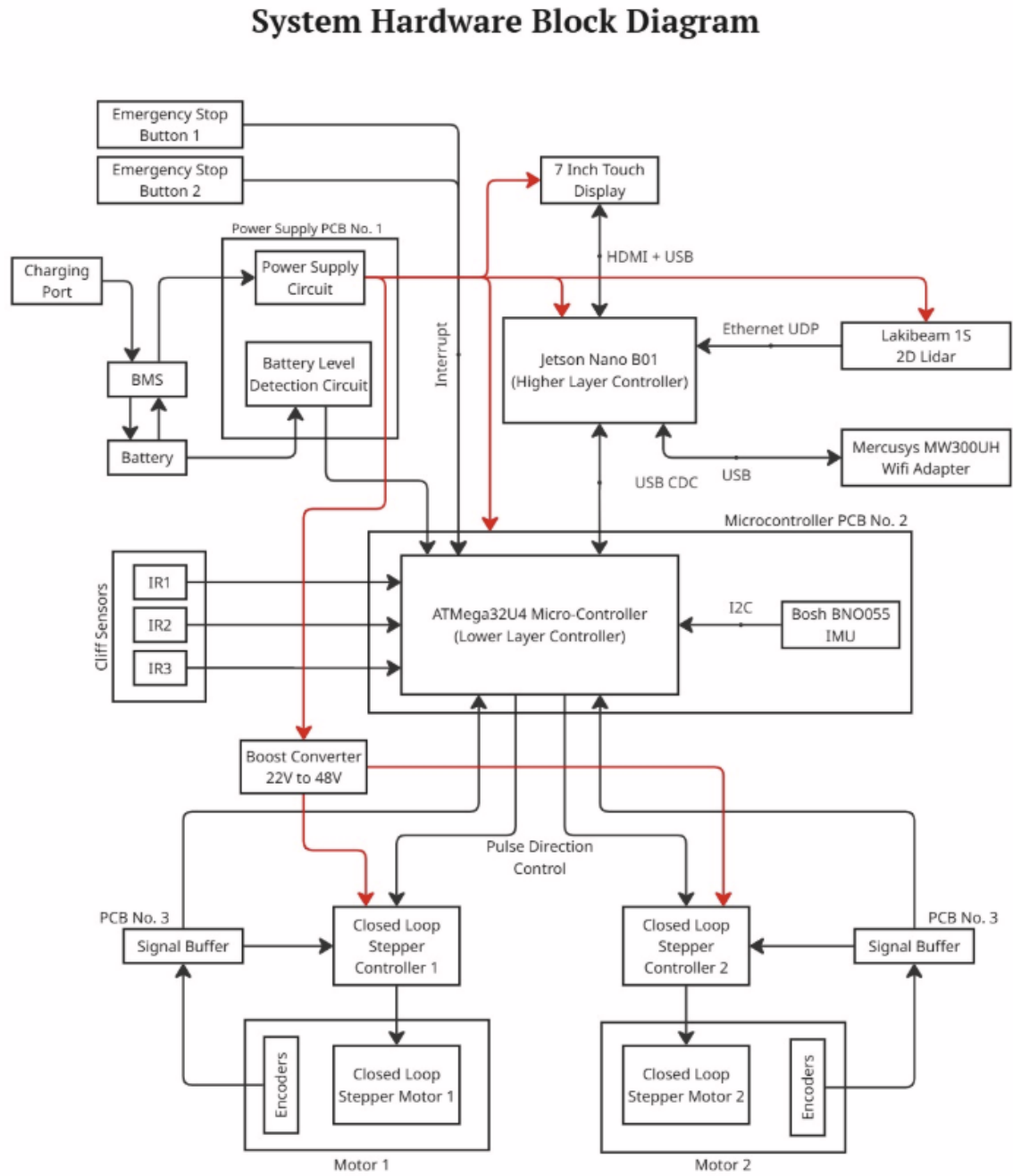


Figure 85: Final System Hardware Block Diagram

After numerous design iterations and rigorous validation cycles, given below is the final system hardware block diagram of our AMR system.

15.2 Power PCB

All power for the control electronics and drive motors is derived from a single 24 V battery input on the PCB. From this bus, a boost converter raises the voltage to 36–40 V and

distributes it to each motor driver. Concurrently, a dedicated buck regulator steps the 24 V down to 5 V to power the Jetson Nano, and that same 5 V rail is further tapped off and regulated down for the microcontroller unit. In a separate branch, a second buck converter drops the 24 V to 12 V to feed the LiDAR sensor.

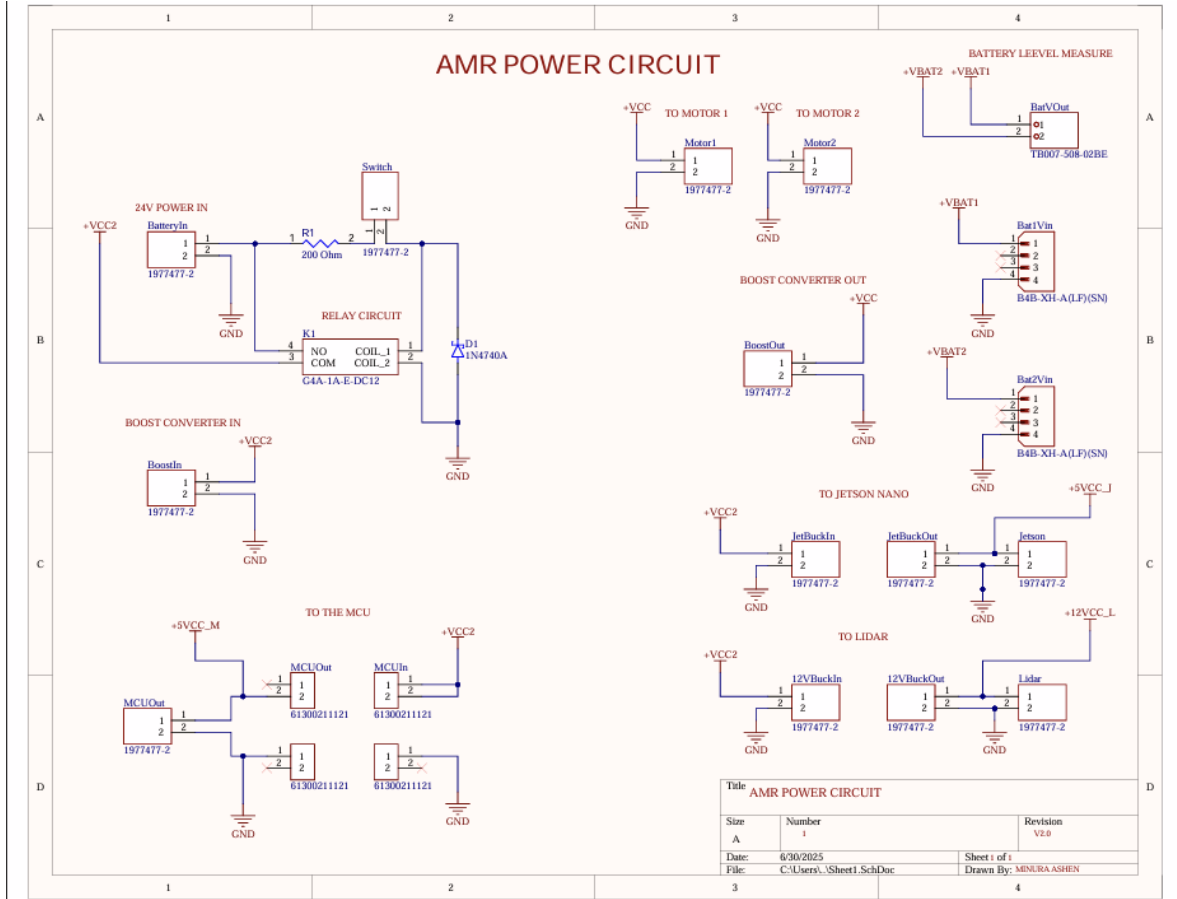


Figure 86: Power Circuit Schematic

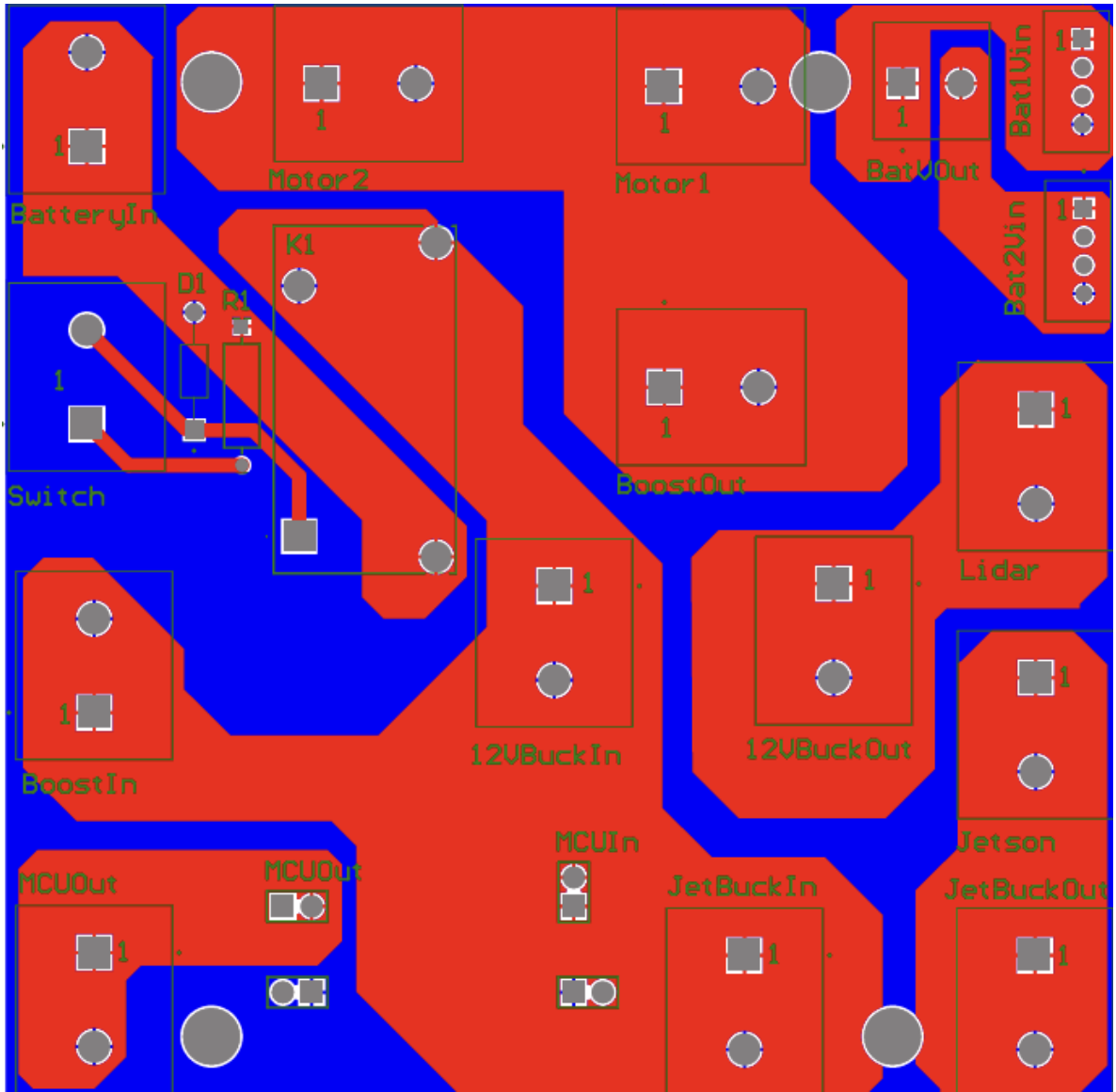


Figure 87: Power Circuit Schematic

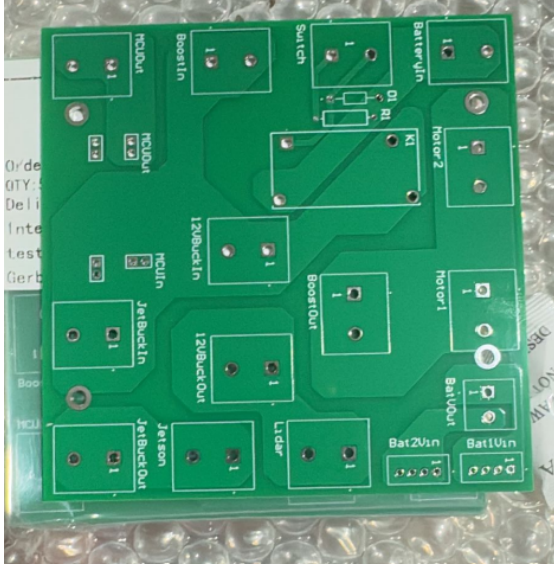


Figure 88: Bare Power PCB

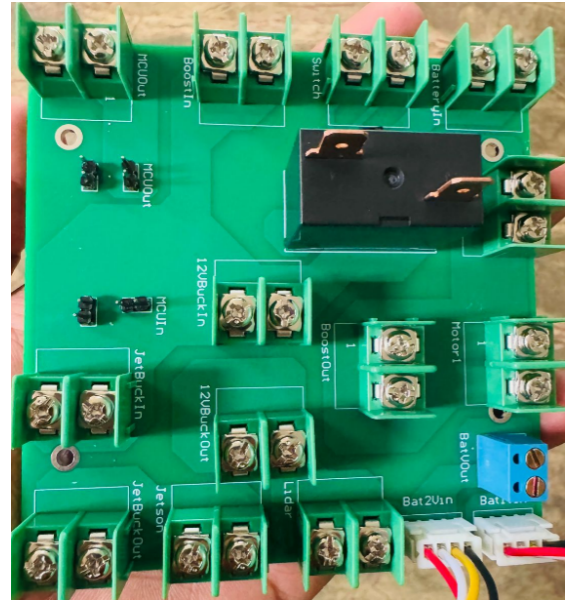


Figure 89: Soldered Power PCB

15.3 Main PCB

This custom controller PCB serves as the central command hub for precise motion control and sensor-driven feedback in our AMR platform. At its heart sits the ATmega32U4 MCU, which orchestrates dual NEMA24 stepper motors via dedicated PUL, DIR, and EN lines while simultaneously processing high-speed differential encoder signals through SN75157DR receivers for accurate position tracking. Power can be supplied either through the USB-C port or an external 5 V input, with an automatic switchover circuit and onboard overcurrent protection ensuring uninterrupted operation. The board also integrates an I²C-connected Adafruit BNO055 IMU for real-time orientation data, three analog cliff sensors for edge detection, and dual voltage dividers for continuous battery monitoring. A comprehensive communication suite—including USB-C for data transfer, an SPI header for firmware updates, and a JTAG interface for in-depth debugging—allows seamless development and integration. Meticulous PCB design practices, such as controlled-impedance differential routing, extensive ground planes, and strategically placed decoupling capacitors, guarantee signal integrity and EMI resilience. Status LEDs, push buttons, and accessible test points round out the feature set, making this board a robust, modular solution for next-generation robotic control and rapid prototyping.

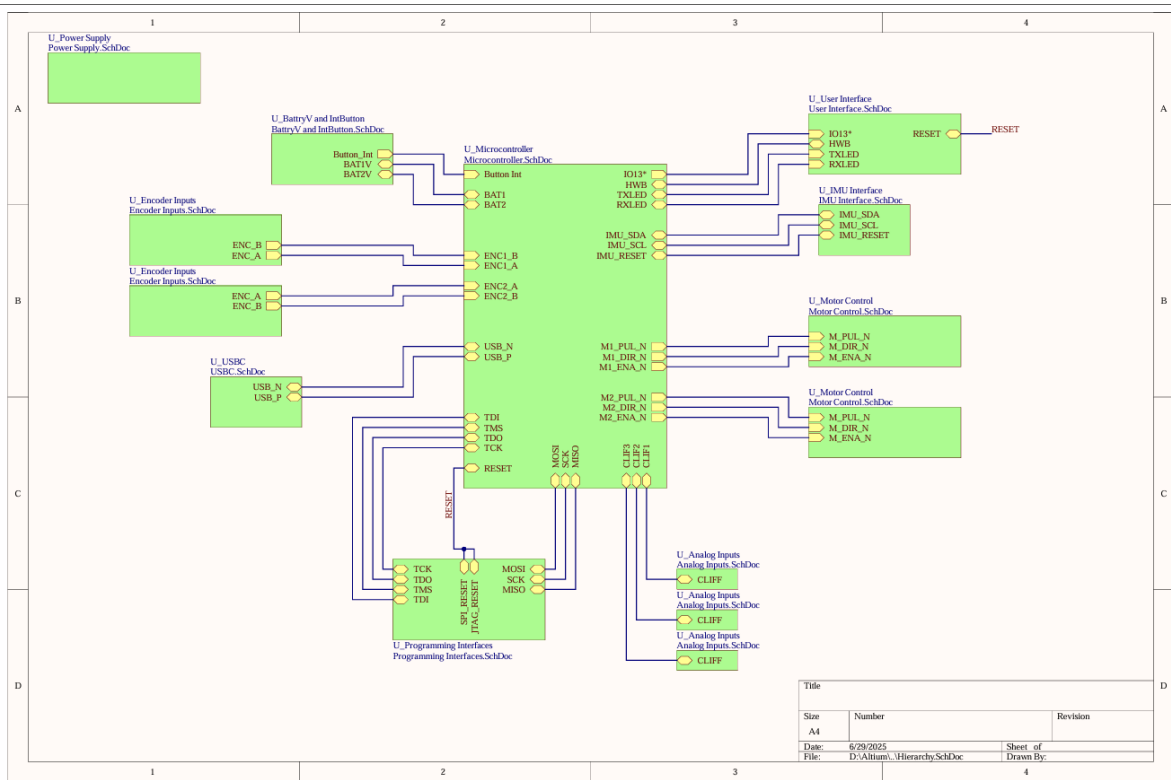


Figure 90: Main PCB Hierarchy

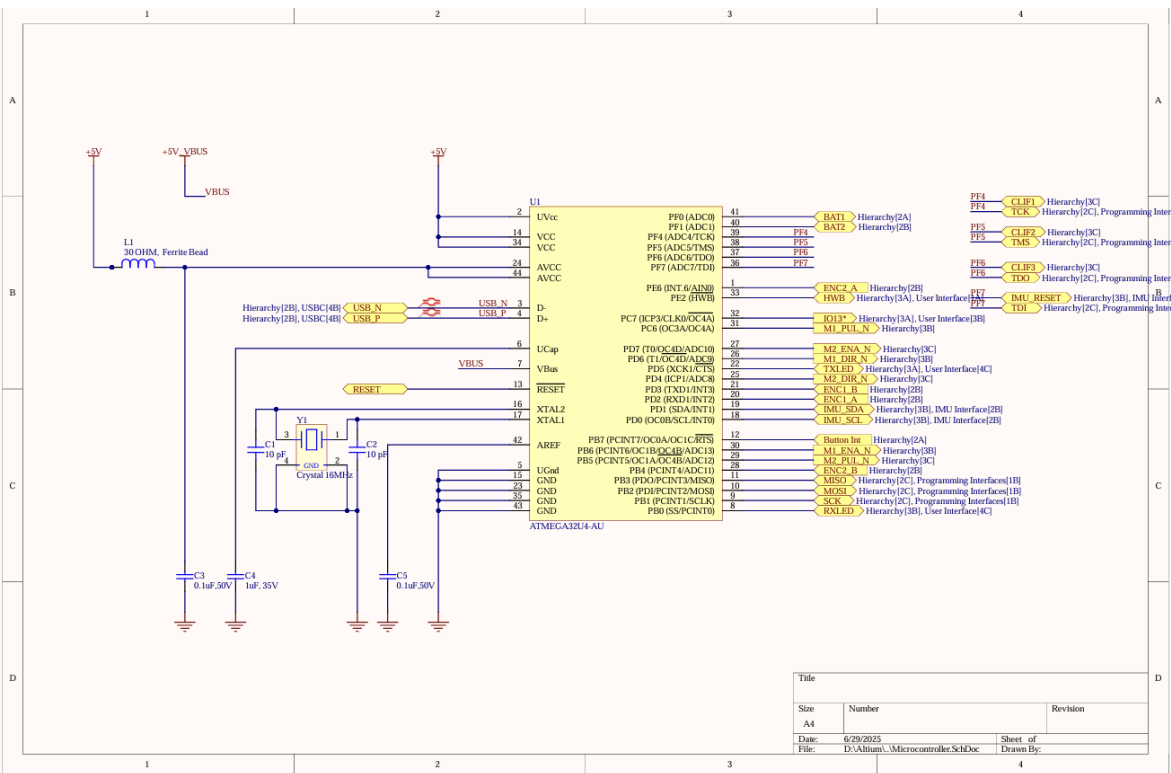


Figure 91: Micro-Controller

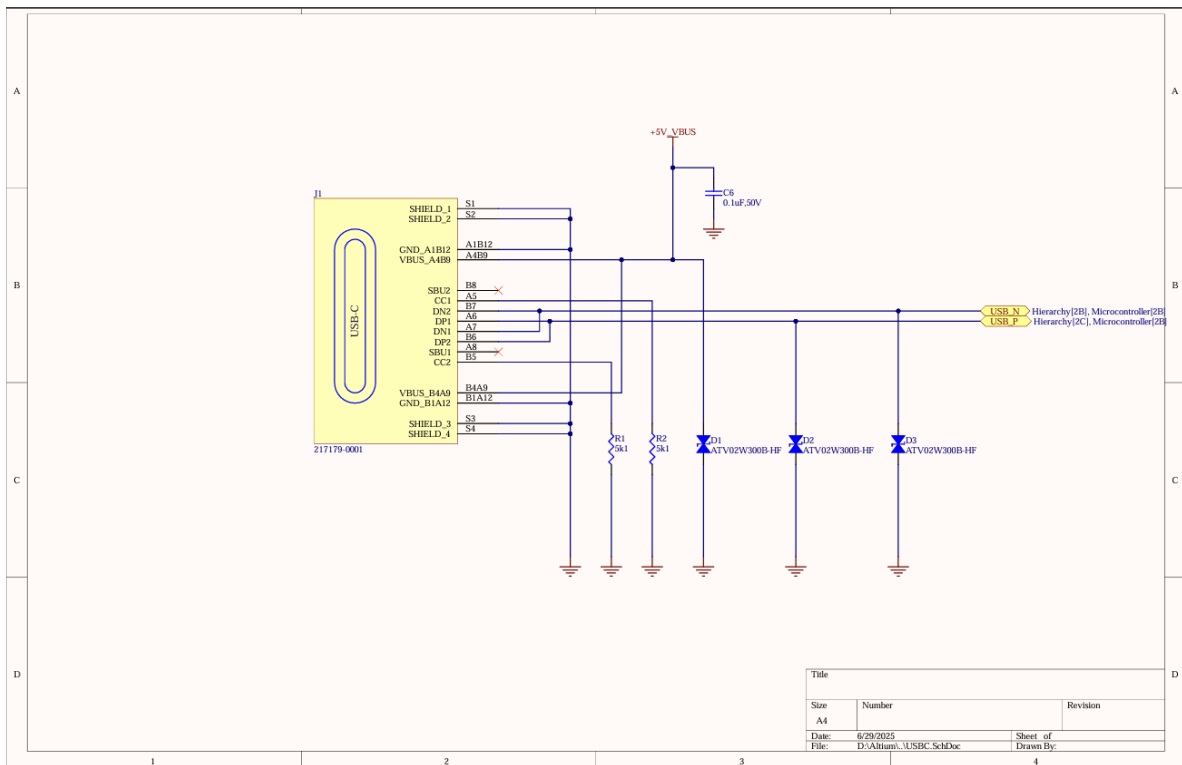


Figure 92: USB-C

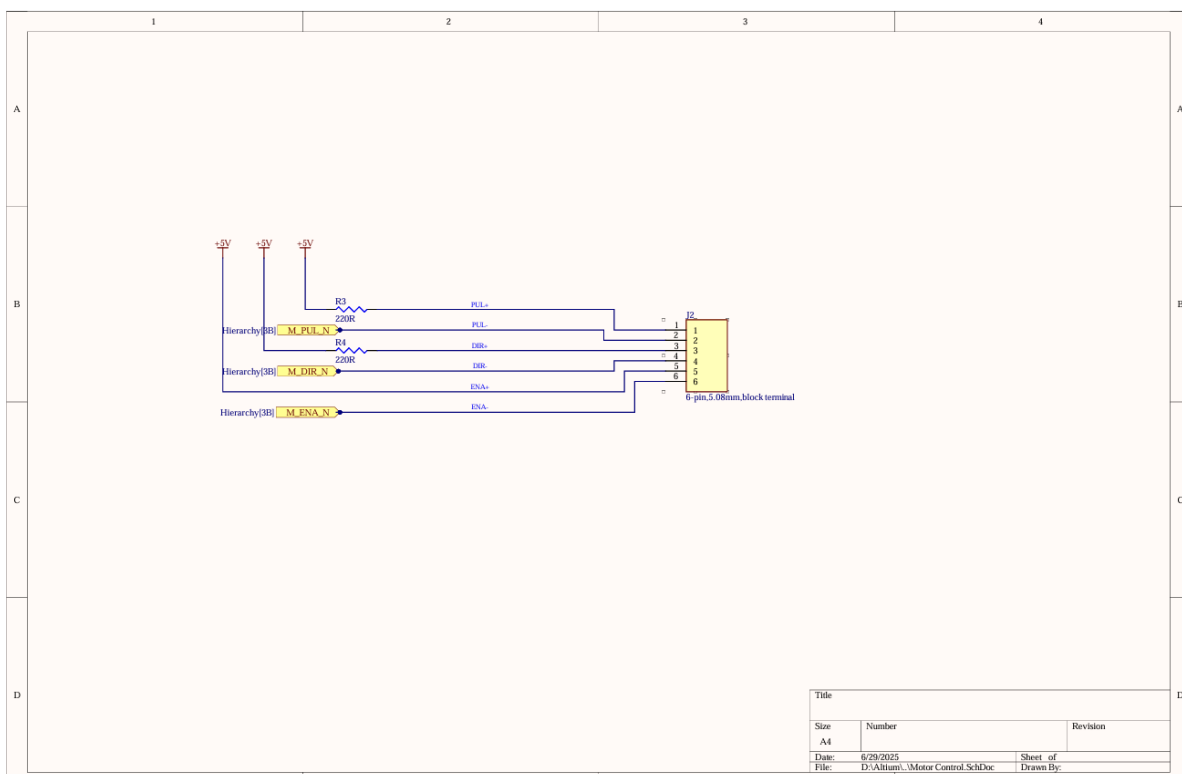


Figure 93: Motor Control

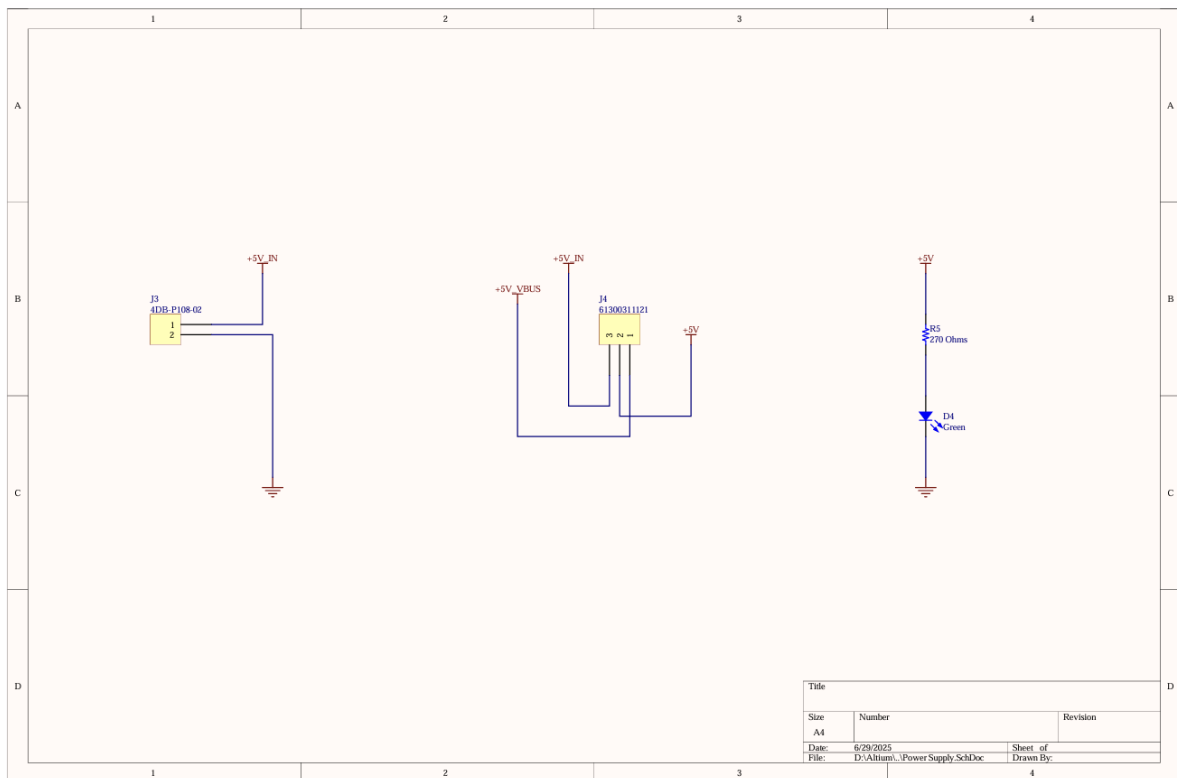


Figure 94: Power Supply

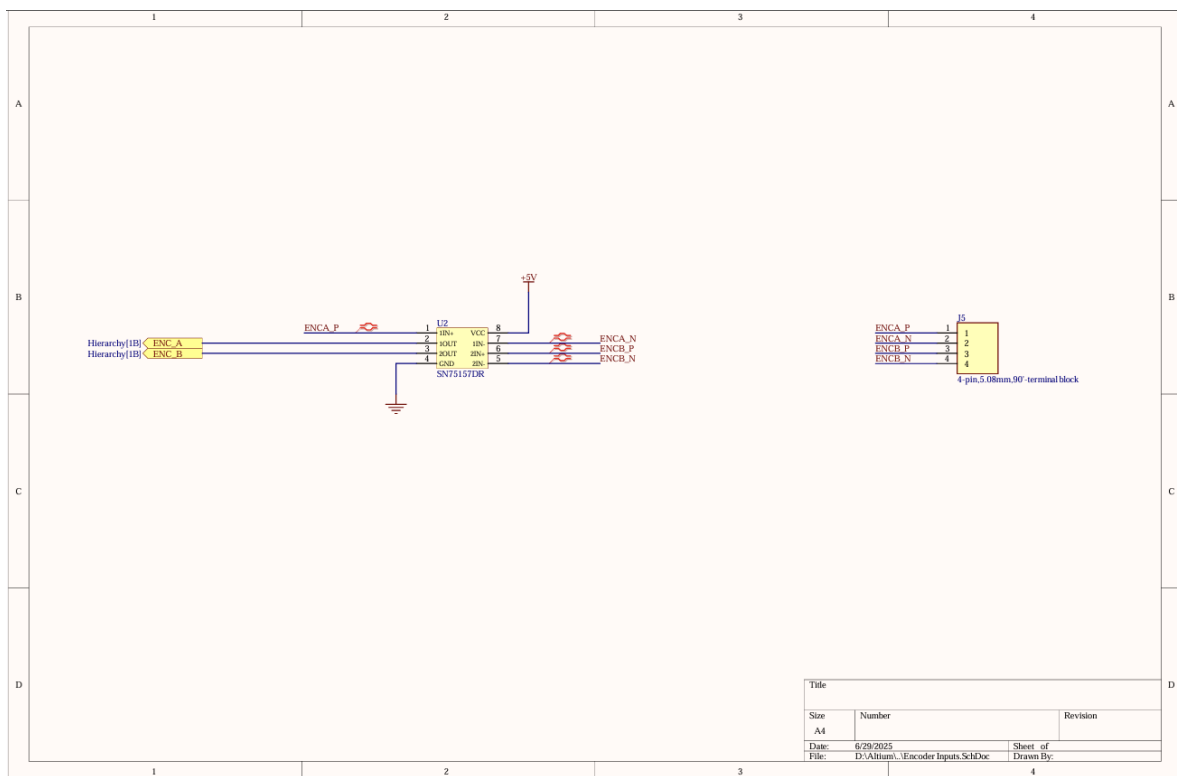


Figure 95: Encoder Inputs

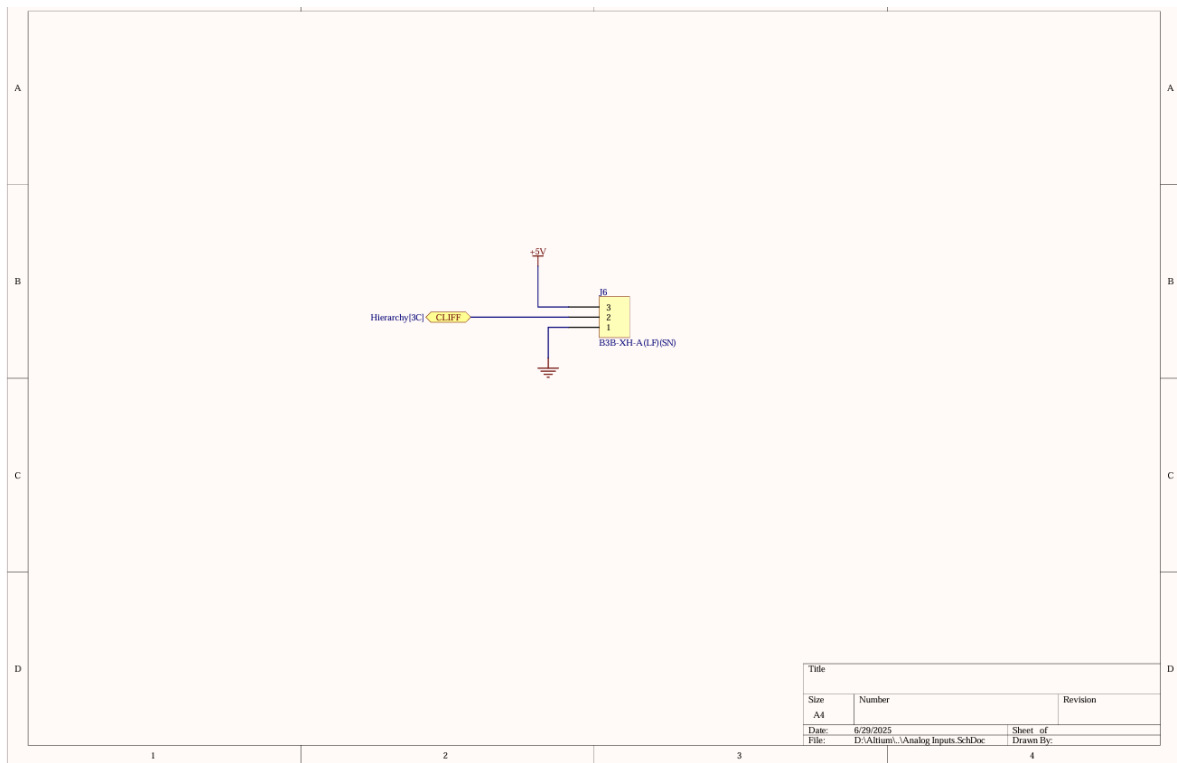


Figure 96: Analog Inputs

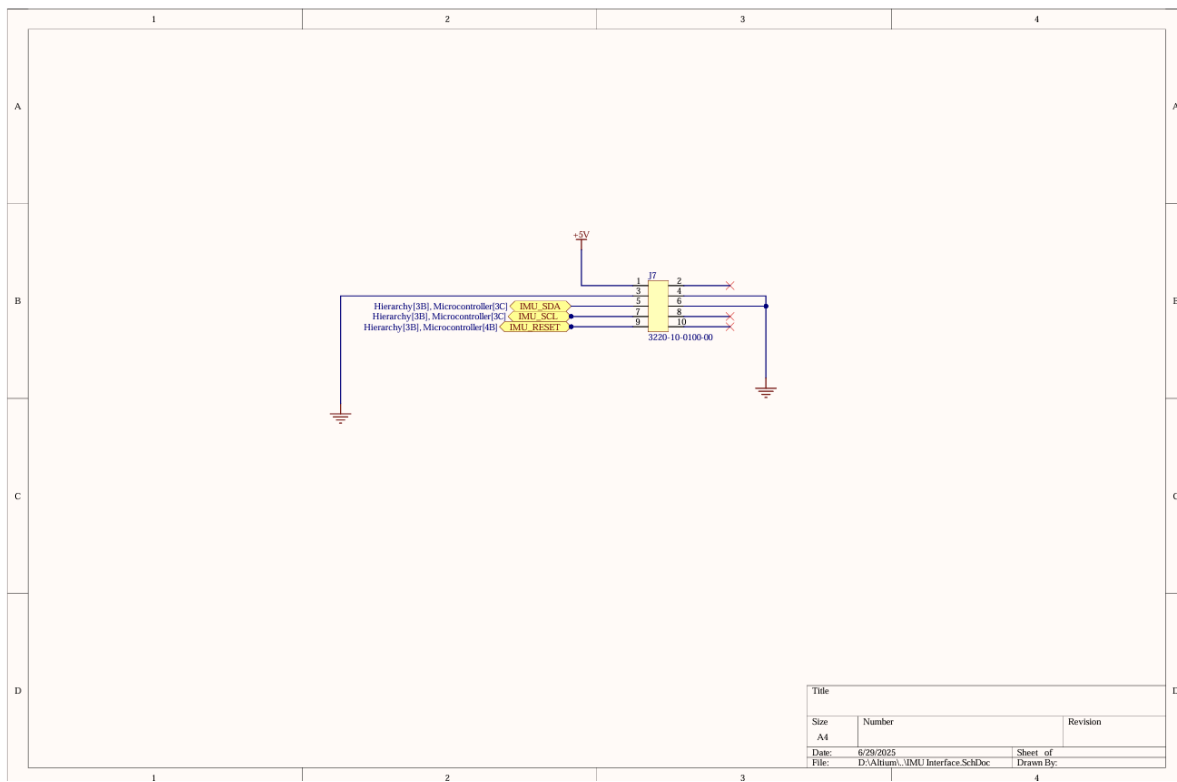


Figure 97: IMU

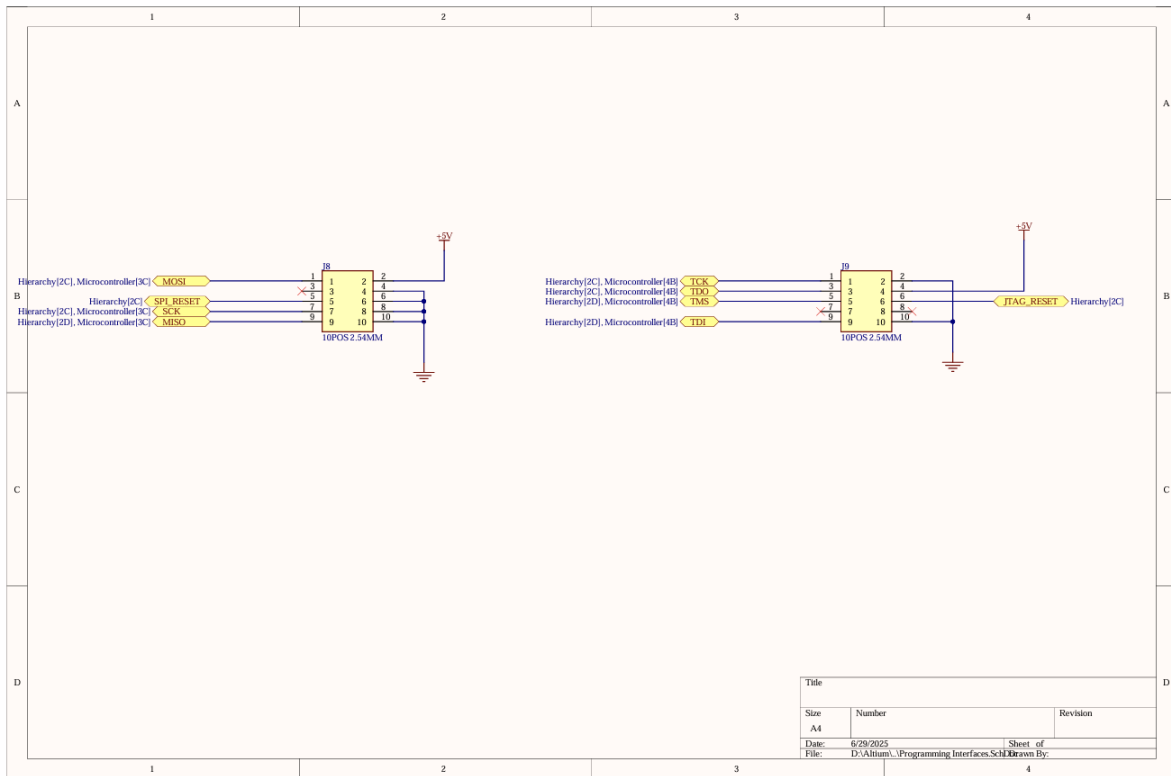


Figure 98: Programming Interfaces

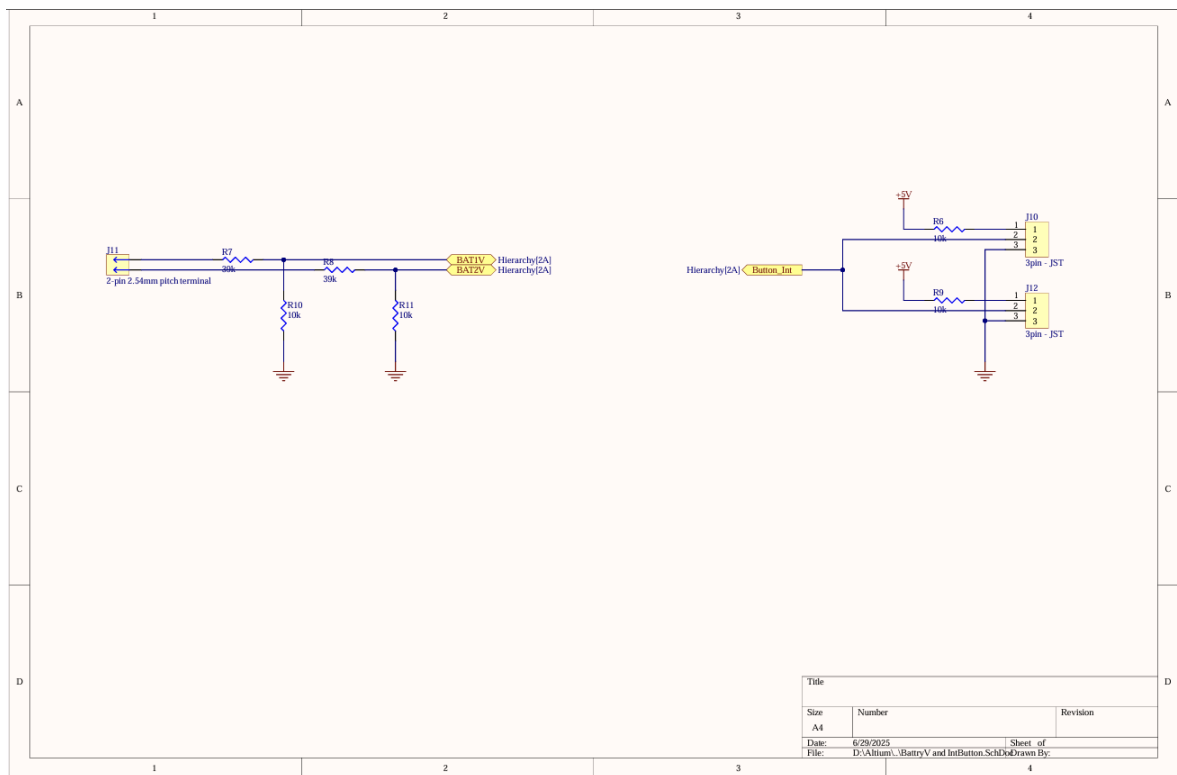


Figure 99: Battery and Int Button

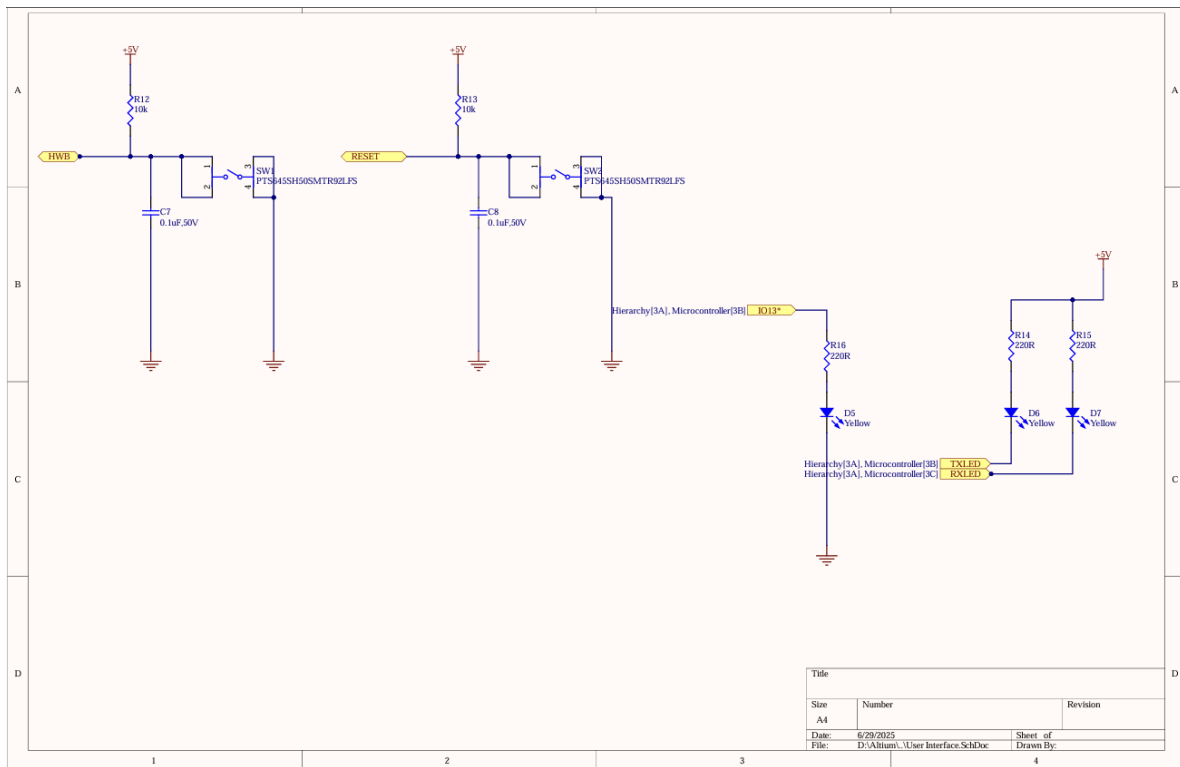


Figure 100: User Interface

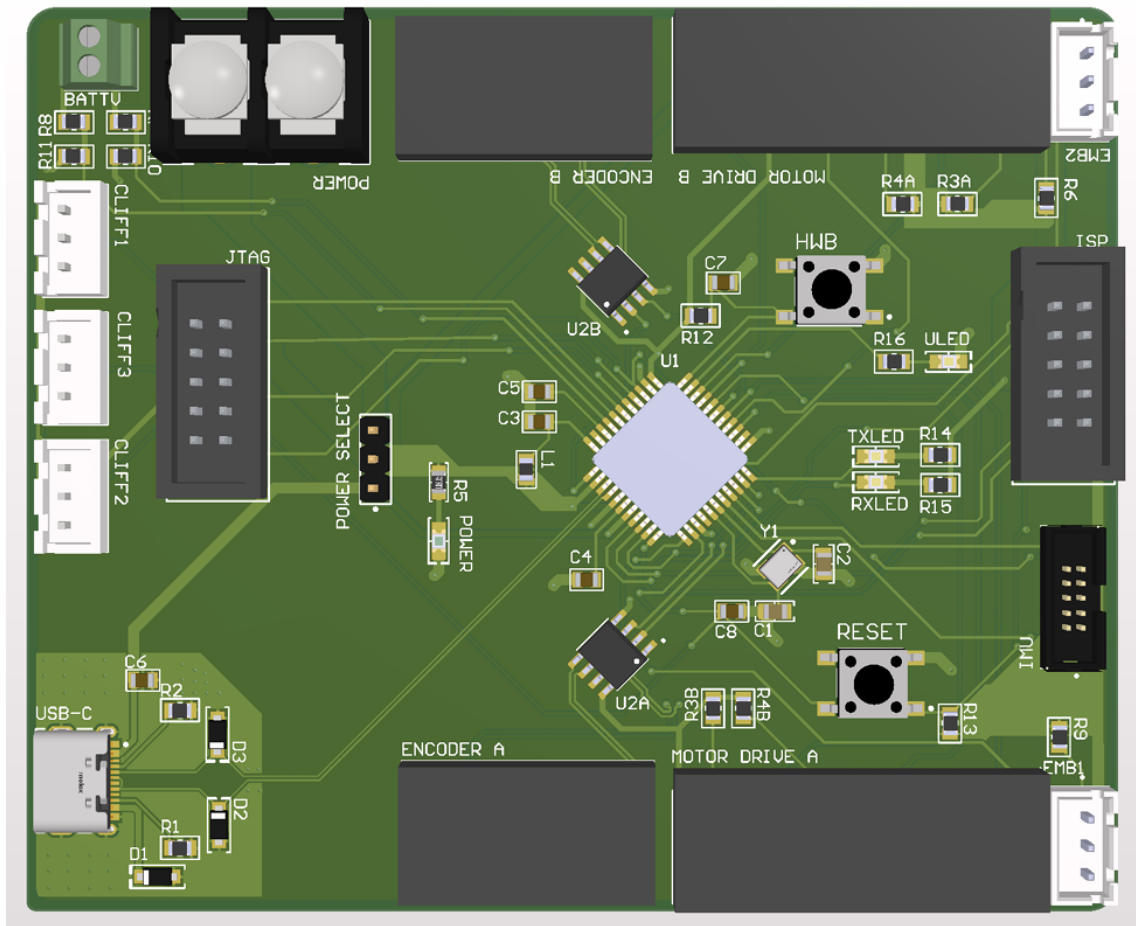


Figure 101: 3D View

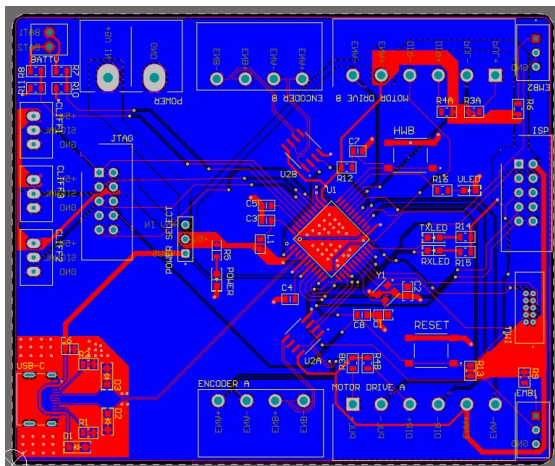


Figure 102: Main PCB Top Layer

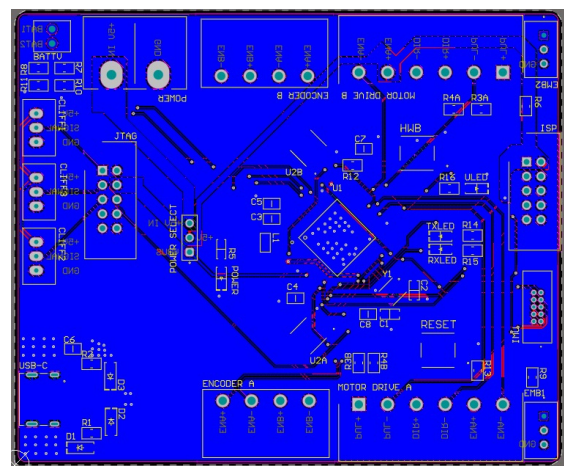


Figure 103: Main PCB Bottom Layer

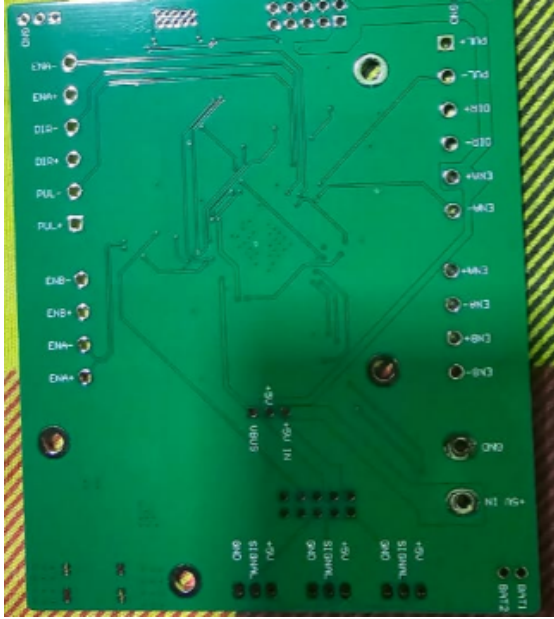


Figure 104: Main PCB

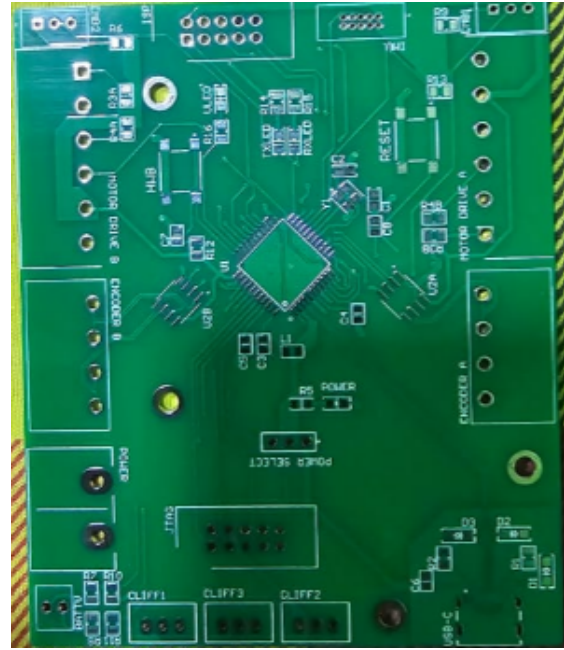


Figure 105: Main PCB

15.4 Enclosure Design Updates

Although the core chassis and enclosure design of our AMR platform have remained unchanged to preserve structural integrity and reliability, we have implemented numerous targeted refinements to support a more modular component layout.

For placement of the pcb and the buffer circuit pcb, an enclosure was created for modular layout and protection.

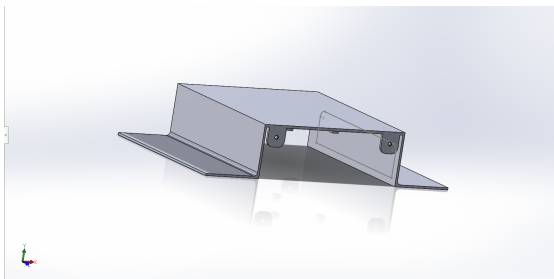


Figure 106: Main and Buffer PCB enclosure Top Part

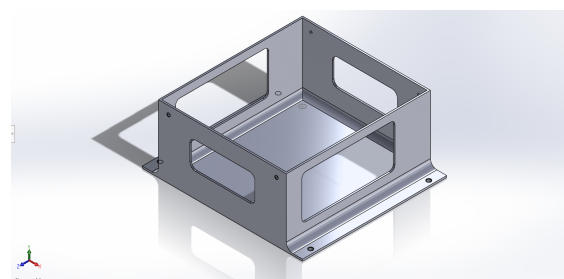


Figure 107: Main and Buffer PCB enclosure Bottom Part

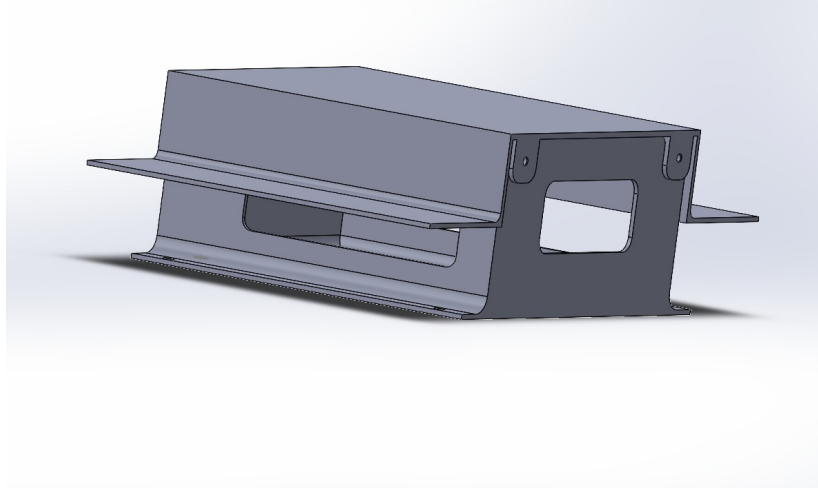


Figure 108: Assembly

Additionally for the placement of the power circuit PCB, an enclosure was created for modular layout and protection.

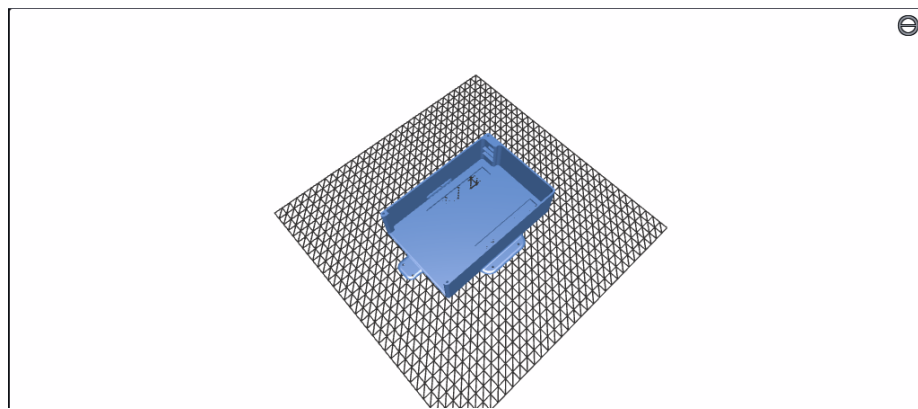


Figure 109: Power PCB enclosure Top Part

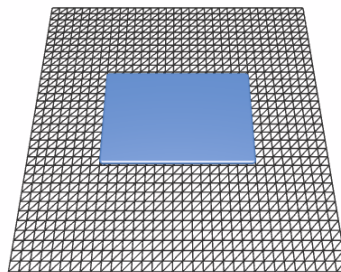


Figure 110: Power PCB enclosure Bottom Part

16 Software Implementation Details

16.1 Lower-Layer Controller (MCU) Overview

Our ATmega32U4-based controller forms the hard-real-time backbone of the AMR, executing a deterministic 10 ms cycle to integrate sensing, estimation, motion profiling, and actuation while streaming telemetry to the Jetson Nano over USB. Firmware is organized into modular drivers—each paired with a header (`.h`) and source (`.c`) file—and all peripheral mappings, timing constants, and robot-specific parameters reside in a single `config.h`. This centralization ensures that any change in wiring or mechanical design requires editing only one file.

Configuration & Abstraction

- System clock set to 16 MHz (F_CPU)
- Pin assignments for stepper drivers, encoders, ADC channels, emergency stop, and I²C lines
- Prescalers and sample counts for timers and ADC

Analog Sensor Interface

- Initializes ADC with AVcc reference, disables digital inputs
- Hardware averaging across multiple samples
- Converts raw counts to millivolts for dual-battery monitoring and normalized voltages for three cliff sensors

IMU (BNO055) Driver

- Low-level TWI primitives with status checks
- GPIO reset and NDOF fusion-mode configuration
- Reads Euler angles, angular rates, linear accelerations, and handles calibration

Encoder & Odometry

- External and pin-change interrupts update quadrature tick counters
- Kinematic conversion of ticks to millimeters and degrees
- Real-time velocity calculation and emergency-stop override

Motion Profiling & Timing

- Trapezoidal velocity profiles via a state-machine API for smooth acceleration/deceleration
- 64-bit microsecond timestamp API from a free-running Timer0 with overflow ISR

10 ms Main Control Loop

1. **Emergency Check:** Halt immediately on panic button
2. **Command Parsing:** Read CSV-formatted instructions (mode flags, targets, teleop) over USB
3. **Sensor & Odometry Update:** Fusion of encoder, IMU, battery, and cliff data
4. **Motion Control:** Autonomous trajectory execution or teleop joystick mapping
5. **Telemetry Streaming:** Send pose, sensor readings, and diagnostics
6. **Shutdown Handling:** Safely disable motors and flush buffers on fault

This architecture delivers precise timing, easy maintainability, and a clear path for future enhancements.

16.2 Software Stack on the Jetson Nano

The Jetson Nano functions as the central processing unit for our AMR, overseeing high-level control loops, integrating sensor data, and executing path planning routines. The codebase is modularized into configuration, communication, sensing, localization, planning, and execution segments, allowing any one component to be modified, tested, or swapped independently without causing cascading changes.

16.2.1 Configuration & Utilities

To ensure consistency and ease of tuning, all compile-time parameters—such as port identifiers, communication rates, EKF (Extended Kalman Filter) covariance values, and map specifications—are declared in a single header file (`config.hpp`). This centralized setup makes adapting to different hardware variants or environments as straightforward as editing one file. Supporting routines live in `robot_utils.hpp`, which houses coordinate transform functions, timestamp generation, and logging macros so that the main algorithmic modules remain uncluttered by boilerplate code.

Global Configuration

- Serial settings matched to the ATmega32U4
- UDP network parameters for telemetry and control separation
- LiDAR driver constants for reliable packet reassembly
- EKF process and measurement variances, exposed for rapid tuning
- Map resolution and inflation radius to decouple the planners from the robot’s footprint

Utility Routines

Key helper functions include coordinate conversions between world and grid indices, angle normalization, high-precision timestamps for uniform logging, and logging macros that automatically include file, line, and function metadata.

16.2.2 Communication Modules

Communication is split into serial (UART) and UDP handlers, each designed for robustness and minimal latency.

Serial Communication Handler

Motor commands and low-level sensor data are controlled by the ATmega32U4 via a C++ interface (`serial_com.cpp/.hpp`). Packet framing, CRC checks, and thread-safe buffering ensure no telemetry is lost, even under bursty loads. If no valid packet arrives within a configurable timeout, the port resets automatically and error counters are exposed for monitoring.

UDP Communication Handler

Using Boost.Asio, non-blocking UDP sockets (`udp_com.cpp/.hpp`) manage real-time telemetry and remote commands. Incoming messages update atomic control variables immediately, while outgoing telemetry frames carry sequence numbers so the UI can detect any packet loss. To maintain throughput, low-priority debug messages are dropped if the send queue grows too large.

16.2.3 Sensor Data Acquisition

LiDAR Data Receiver

The LakiBeam LiDAR streams packets at up to 20 Hz. A low-level driver stitches 360° scans using packet counters and a double-buffer pattern. A high-level handler filters invalid points, converts them to metric tuples, and exposes them via a thread-safe getter; an optional CSV dump aids offline SLAM tuning.

Serial-Stream Sensor Parser

Wheel encoder deltas and IMU samples arrive over UART, where they are decoded, bias-corrected, and flagged if stale or duplicate to prevent EKF drift.

16.2.4 Localization & Sensor Fusion

Extended Kalman Filter (EKF)

A discrete EKF fuses wheel odometry, gyro, accelerometer, and optional LiDAR odometry into a 5D state $(x, y, \theta, v, \omega)$. Differential-drive kinematics predict the pose, then measurement updates correct drift; automatic zero-velocity resets collapse speed uncertainty, and any sensor dropout forces its Kalman gain to zero.

Localization Orchestration

`RobotLocalization.cpp` initializes the filter and runs a 100 Hz loop for predict/update in sync with sensor reads. On failure (e.g., LiDAR dropout), it switches to dead-reckoning with inflated covariance and logs timestamped state and covariance diagonals when enabled.

16.2.5 Path Planning

C++ Implementation (A* & PRM)

Both grid-based A* and sampling-based PRM (Probabilistic Roadmap) planners live in `maincp.cpp`. After thresholding and inflating the map by the robot radius, A* performs an eight-connected search with Euclidean cost and heuristic, while PRM samples points and connects neighbors within a radius R . The shorter path is chosen, and waypoints with heading changes $> 10^\circ$ are flagged for profiling.

Visualization

To illustrate the effectiveness of our two planning approaches, we overlay the computed trajectories on the same occupancy map.

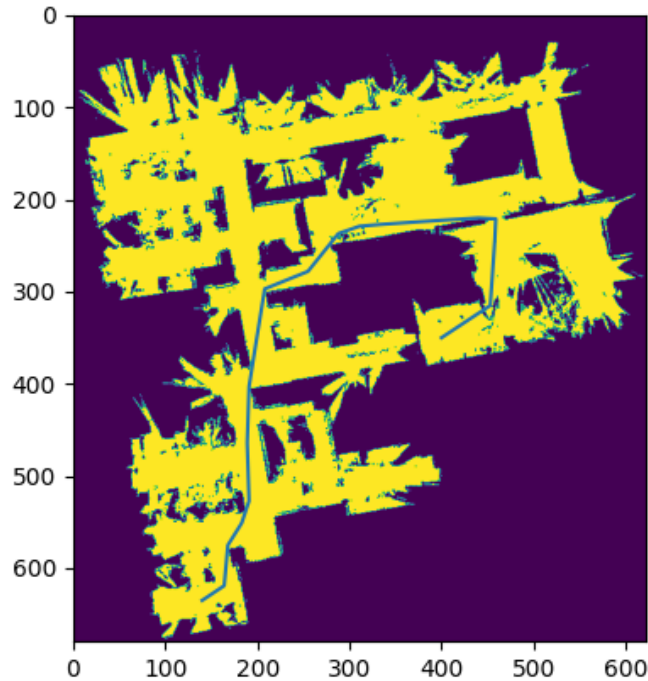


Figure 111: A* Planned Path overlaid on the inflated occupancy grid. The yellow line follows the shortest grid-based route, hugging the free-space boundary and making precise 45° and 90° turns at corridor junctions.

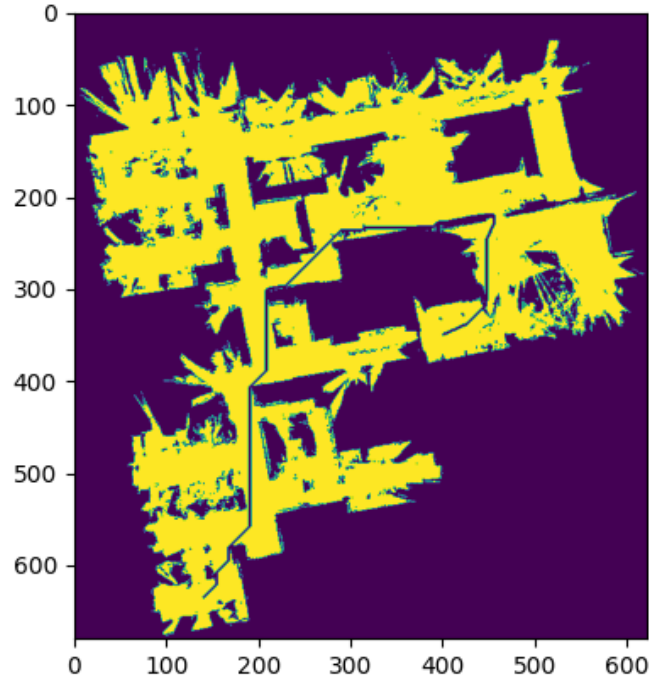


Figure 112: PRM Planned Path on the same map. Randomly sampled waypoints yield longer straight segments in open areas, with turns only at connection nodes.

In Figure 111, the A* algorithm searches cell-by-cell across an 8-connected grid. Starting from the bottom-left, it maintains one-cell clearance from walls, navigates narrow passages, and guarantees minimal path length under the grid metric.

Figure 112 shows the Probabilistic Roadmap solution. A sparse set of samples is connected into a graph, then searched to produce a path that:

- Cuts more directly across large open areas via straight-line connections.
- Introduces fewer, but sometimes longer, segments compared to the grid-constrained A* path.
- Matches A* performance in tight corridors, yet yields smoother trajectories in sparse regions.

By comparing these visualizations, we see that A* guarantees the shortest grid-based route, while PRM exploits open spaces for more natural "as-the-crow-flies" segments.

To further demonstrate the robustness of our planning algorithms, Figures 113 and 114 overlay a different start-goal pair on the same occupancy map.

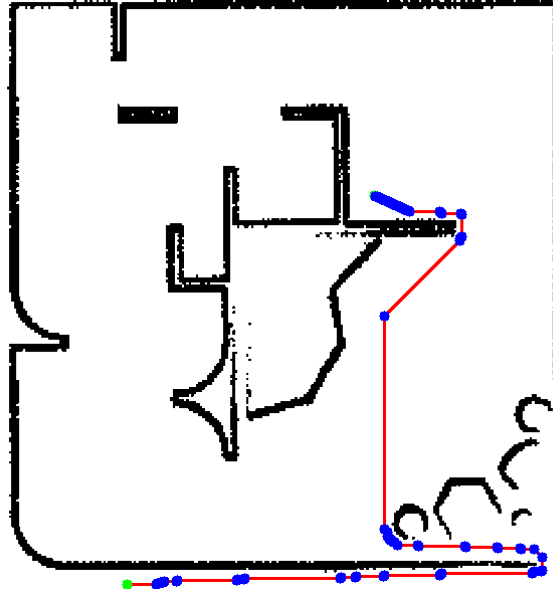


Figure 113: A* planned path for an alternate route. The A* algorithm again searches cell-by-cell on the 8-connected grid, maintaining one-cell clearance from walls, navigating narrow corridors with precise 45° and 90° turns, and guaranteeing the shortest possible route under the discrete grid metric.

In Figure 113, the A* planner begins at the green start marker and systematically expands through adjacent free cells. Its tightly zig-zagging red polyline reflects the grid's connectivity constraints, ensuring that every move is optimal in terms of grid distance, even in complex, cluttered sections of the map.

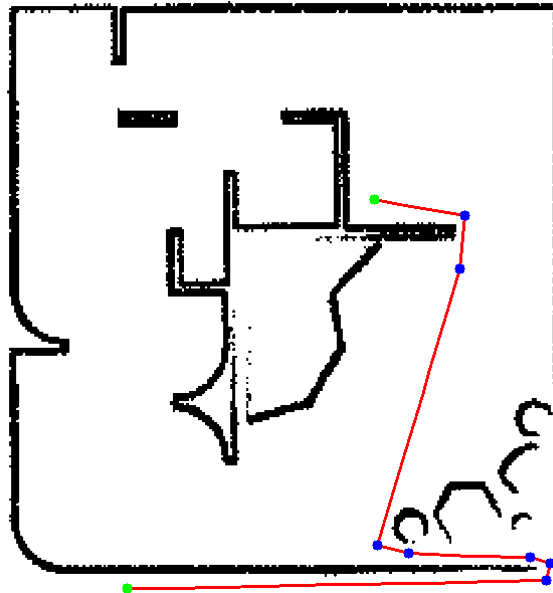


Figure 114: PRM planned path for the same alternate scenario. The Probabilistic Roadmap samples free-space nodes and connects them with straight-line edges, producing longer, direct segments in open areas and fewer waypoints in tight passages.

Figure 114 shows the PRM solution for this new start-goal configuration. By sprinkling samples across the free space and linking neighbors, PRM constructs a graph whose

shortest-path search yields a red trajectory with smoother, "as-the-crow-flies" edges. It retains corridor-following behavior where necessary but dramatically reduces the number of turns and stop-start motions compared to A*.

Comparing Figures 113 and 114 highlights the trade-off between guaranteed grid-optimality and smooth, continuous trajectories. While A* ensures the minimal cell-based route, PRM delivers a more natural path with fewer abrupt heading changes, improving kinematic feasibility and average speed in environments combining narrow passages and open areas.

Python Reference Implementation

A prototype in `main.py` uses NumPy, NetworkX, and Matplotlib to binarize/inflate maps, build PRM graphs, and plot both planners with turning-point overlays.

- **Binarization and Inflation:** Transforms and expands the map for streamlined processing.
- **PRM Graph Construction:** Employs NetworkX to create and query Probabilistic Roadmap (PRM) graphs for path planning.
- **Visualization:** Renders the occupancy grid with the final path and turning points overlaid using Matplotlib.

16.2.6 Mapping (Simultaneous Localization and Mapping)

Two complementary SLAM strategies are implemented: an Online SLAM for real-time, low-latency mapping within the control loop, and an Offline SLAM pipeline for batch processing with global optimization. Both utilize logged LiDAR scans and pose estimates but vary in computational demands, consistency, and output quality.

Online SLAM

Online SLAM maintains a lightweight occupancy grid, updated at 10 Hz within a 10 ms control loop, enabling reactive obstacle avoidance and live telemetry for remote dashboards. Each cycle involves:

- **Pose Retrieval:** Fetches the latest pose from the Extended Kalman Filter (EKF).
- **Ray-Tracing:** Applies Bresenham's algorithm to update the grid:
 - *Free Cells:* Marks cells along LiDAR beams as free (misses).
 - *Occupied Cells:* Marks beam endpoints as occupied (hits).
- **Log-Odds Updates:** Increments grid cells with bounded log-odds to avoid numerical overflow.
- **Optional Filtering:** Uses a fast median filter to reduce isolated spurious hits.

Key Attributes:

- **Latency:** Updates complete in milliseconds, suitable for real-time collision checks.
- **Resource Usage:** Fixed-size grid and simple ray-tracing ensure predictable CPU and memory usage.
- **Drift:** Lacks loop closures, causing odometry errors to accumulate over time.

Offline SLAM

Offline SLAM refines robot poses and reconstructs the occupancy grid with loop-closure and global smoothing for high-fidelity post-mission maps. The process includes:

1. **Initial Mapping:** Inserts logged scans into an OccupancyMap using Bresenham’s algorithm for hit/miss updates.
2. **Pose Refinement:** Uses a ScanMatcher for hierarchical grid search to align scans with the current map.
3. **Trajectory Storage:** Stores refined poses and timestamps for batch optimization.
4. **Batch Optimization:**
 - *Re-refinement:* Re-aligns scans to the evolving map with weighted corrections.
 - *Loop Closure Detection:* Identifies revisit pairs within a distance threshold and applies pose graph corrections.
 - *Global Smoothing:* Applies windowed averaging to minimize residual jitter while preserving structure.
5. **Final Map Rebuild:** Clears the map, re-inserts scans at optimized poses, applies median filtering, and exports to PGM format.

Key Attributes:

- **Consistency:** Loop-closure and pose-graph smoothing eliminate long-term drift.
- **Fidelity:** Sigmoid-style probability updates and median filtering produce clear obstacle boundaries.
- **Compute Cost:** Multi-level scan matching and global optimization may take minutes to hours based on dataset size.

Comparison of Online and Offline SLAM

Characteristic	Online SLAM	Offline SLAM
Latency	Real-time (sub-10 ms updates)	Batch (minutes to hours)
Consistency	Local; drift over time	Global via loop-closure
Fidelity	Moderate; no global smoothing	High; sigmoid updates + median filter
Compute Load	Low; simple ray-tracing	High; multi-level matching + optimization
Memory Foot-print	Fixed grid buffer	Large; stores all scans and poses
Use-Case	Reactive navigation, live dashboards	Detailed mapping, post-mission review

Table 3: Comparison of Online and Offline SLAM

Online SLAM prioritizes robot safety and responsiveness, while Offline SLAM delivers a polished, drift-free map for documentation, analysis, and advanced planning. Online

mapping supports real-time collision avoidance, whereas Offline SLAM generates a high-fidelity environment model.

16.2.7 Main Control Loop

Application Entry

The main control loop runs on a 10 ms cycle, coordinating all system modules:

1. **Sensor Reading:** Acquires data from UART and LiDAR scans.
2. **Pose Estimation:** Updates the EKF for precise pose estimation.
3. **Path Planning:** Initiates planning for new goals and caches waypoints.
4. **Velocity Commands:** Transmits velocity instructions via UART.
5. **Telemetry Broadcast:** Sends sequenced telemetry data over UDP.
6. **Health Monitoring:** Detects communication timeouts, CRC errors, or missed frames, triggering safe-stop or fallback measures if needed.

Upon shutdown, the loop ensures a clean exit:

- All threads are joined.
- Communication ports are closed.
- Logs are flushed to disk.

On shutdown, all threads join, ports close, and logs flush to disk.

16.3 Remote Monitoring and Control Interface

The Remote Monitoring and Control Interface provides both a rich graphical dashboard and a lightweight command-line tool for real-time observation and teleoperation of the AMR over UDP. These two components: `interface.py` and `udp.py` cater to different user needs, whether you require detailed plots and parameter tuning or a minimal terminal interface.

16.3.1 Graphical User Interface

The GUI, built with PyQt5 and pyqtgraph, offers a comprehensive window where telemetry is visualized, controls are adjusted on the fly, and system logs are monitored.

Overview

A standalone Python application that:

- Receives UDP packets containing sensor readings, debug info, and velocity data.
- Displays live numeric values (speed, angular rate, wheel speeds) in labeled fields.
- Plots time series of key variables (SPD, ROT_SPD, LFT_FDSPD, etc.) with checkboxes to toggle series visibility.

- Provides input fields and "Send" buttons to adjust PID gains (FWD_KP, FWD_KI, ...) and transmit them over UDP.
- Includes Connect/Stop buttons to manage the background UDP listener thread.
- Shows raw and parsed messages, error notices, and acknowledgements in a scrolling log box.

Key Features

- **Threaded UDP Reception:** A dedicated `UDPReaderThread` listens in blocking mode and emits Qt signals for thread-safe UI updates.
- **Real-Time Plotting:** Uses `pyqtgraph` to update plots at high frame rates, automatically trimming buffers to a fixed length.
- **Parameter Validation:** Ensures invalid or empty inputs generate clear log messages rather than silent failures.
- **Graceful Shutdown:** Cleanly stops the UDP thread and closes the application on exit or error.

The User Interface for our AMR delivers a polished, real-time control and monitoring experience through a PyQt5-powered dashboard. Upon launch, users are greeted by a fullscreen splash screen before transitioning to a modular control panel that supports live sensor feedback, error logging, and command dispatching. Communication with the robot is handled via an integrated WebSocket server (`barrier_ws_server.py`), while `listen.py` runs in the background to ingest telemetry and status updates. The interface's flexible architecture—defined across `main.py`, `dashboard.py`, and `splash_screen.py`—facilitates rapid extension for integration, enabling mapping overlays, logging panels, and custom controls with minimal effort.

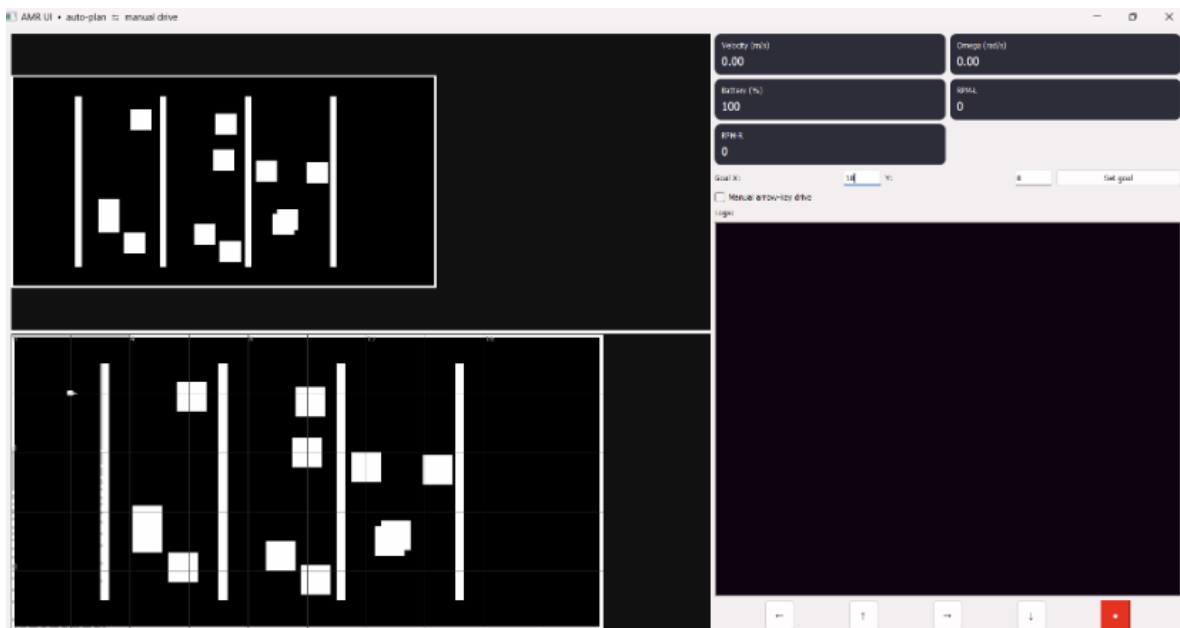


Figure 115: Main Interface

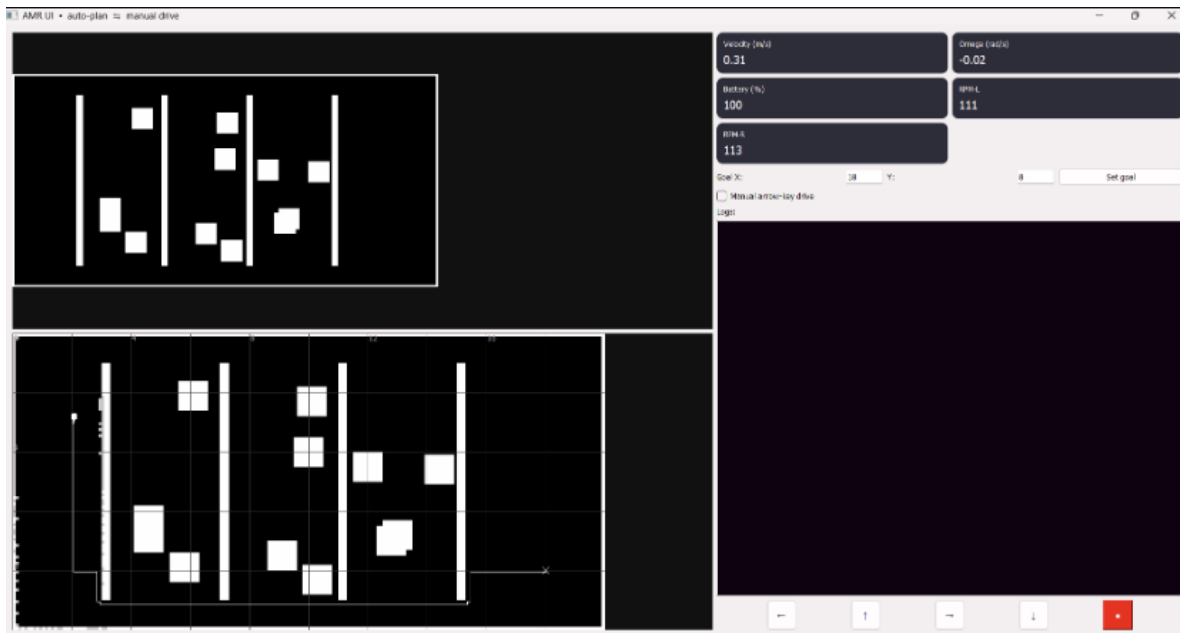


Figure 116: Interface with Mapping

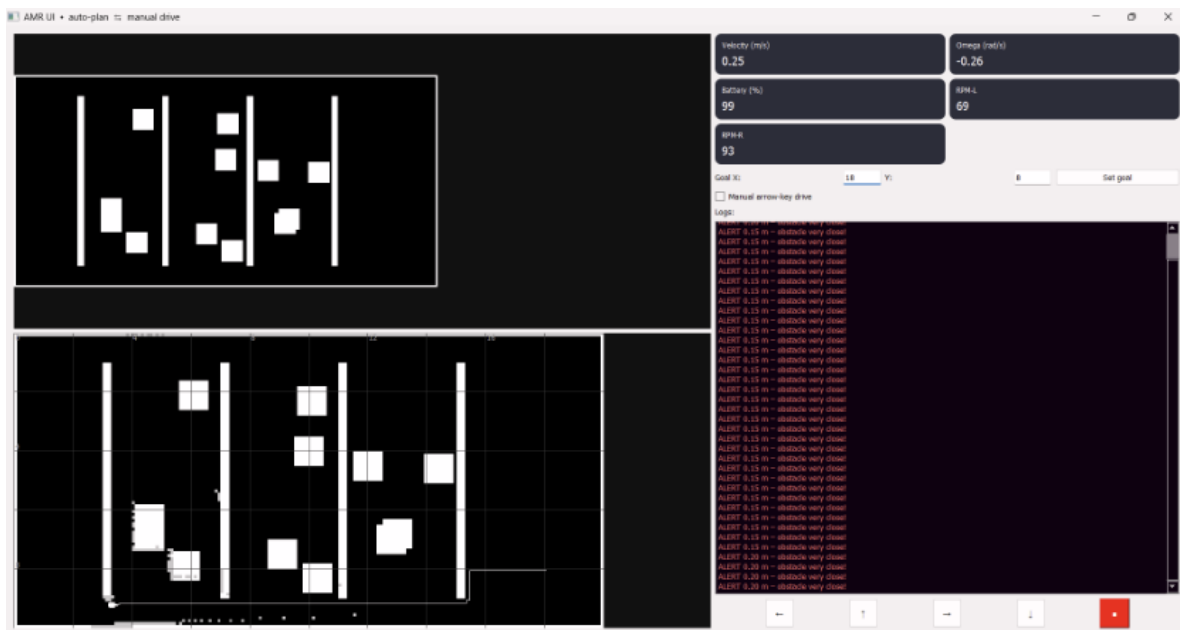


Figure 117: Warning Log Panel



Figure 118: Welcome Screens

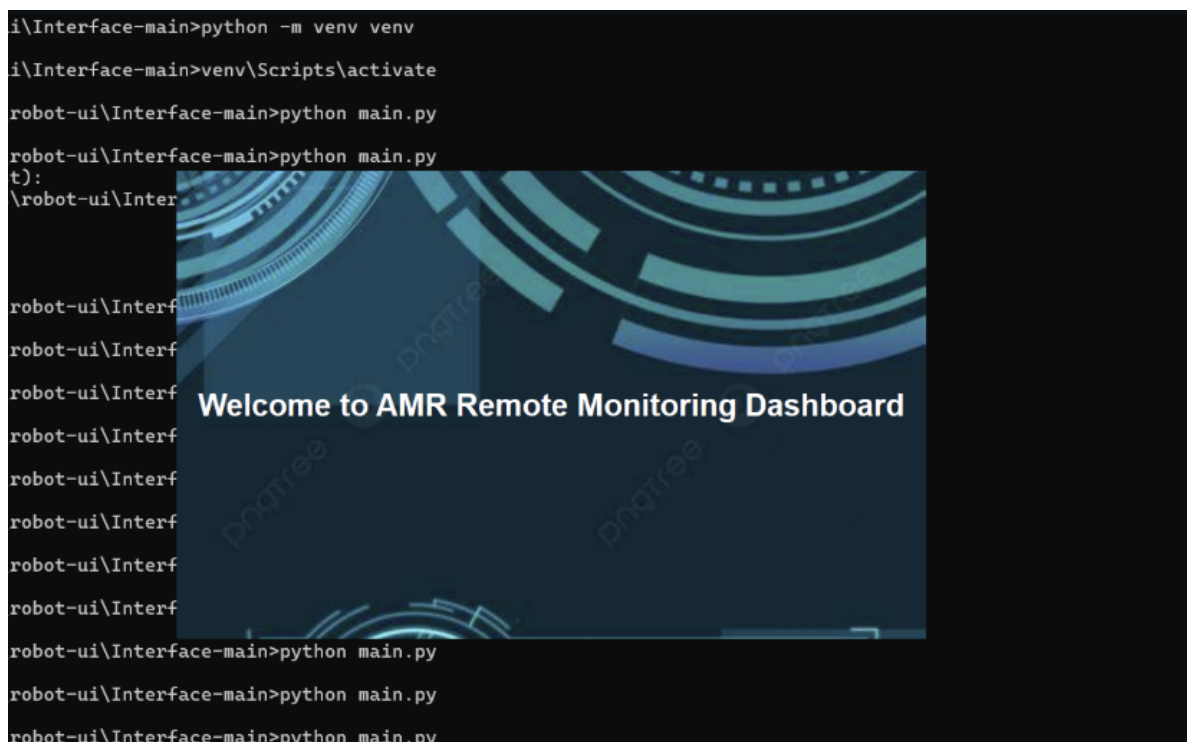


Figure 119: Welcome Screens

16.3.2 Command-Line Interface

For scenarios where a GUI is impractical, the CLI tool provides direct keyboard teleoperation and real-time telemetry display in the terminal.

Overview

A Python script that:

- Binds a UDP socket to receive and print all incoming robot messages with timestamps.
- Switches the terminal to raw mode to capture single-key presses (no Enter required).
- Maps W/A/S/D keys to forward, left, backward, and right commands, sending TELEOP,f,b,l,r UDP packets.
- Monitors the socket and stdin via `select()`, ensuring responsive handling of both telemetry and keypresses.
- Restores original terminal settings on exit (keypress Q or Ctrl-C).

Key Features

- **Non-Blocking I/O:** Uses `select()` to multiplex UDP and keyboard without threading.
- **Terminal Management:** Employs `termios` and `tty` for raw mode and registers an `atexit` handler for safe cleanup.
- **Immediate Feedback:** Prefixes sent commands with [SEND] and prints incoming packets with a timestamp and sender address.

17 Final Product

LIDAR and Wi-Fi modules are mounted on the top deck, each with bespoke brackets matching their footprint.

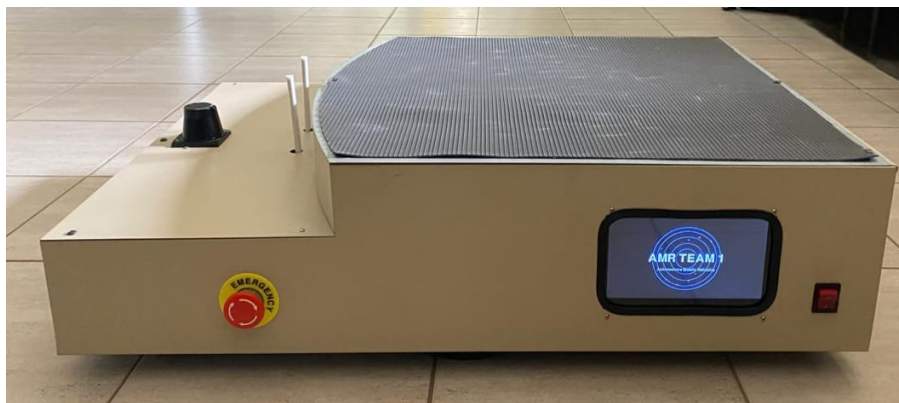


Figure 120: Side panels



Figure 121: LIDAR and Wi-Fi module assembly on the top deck



Figure 122: Completed AMR: fully assembled and ready for deployment

18 Conclusion

In summary, this design documentation has presented a comprehensive, end-to-end exploration of the development process for a cost-effective, modular Autonomous Mobile Robot (AMR) platform. Beginning with an analysis of existing market offerings (Omron, MiR, Geek+) and culminating in detailed hardware schematics, PCB layouts, software

architectures, and enclosure refinements, we have methodically justified every key decision against functional requirements, performance metrics, cost constraints, and user needs.

Throughout the component selection phase, the ATmega32U4 MCU was chosen for its native USB support, ample GPIO count, and strong community resources—providing a lightweight yet capable lower-layer controller. The Jetson Nano SBC brings GPU-accelerated compute power to our system, enabling real-time SLAM, sensor fusion, and advanced path-planning routines. The Lakibeam 1S LIDAR strikes an optimal balance between range (0.1–15 m), angular resolution, and sampling rate, ensuring robust 2D mapping in structured indoor environments. Mechanically, the NEMA 24 closed-loop stepper motors with a 10:1 planetary gearbox deliver the torque, precision (2.4 mm per step), and acceleration ($>5.8 \text{ m/s}^2$) required to maneuver a 25 kg platform reliably. Complementary subsystems—including the Bosch BNO055 IMU, high-gain Wi-Fi adapter, and a capacitive-touch display—complete the sensory and user-interaction suite.

Power management was addressed via a family of DC-DC converters: buck modules to feed the Jetson Nano (5 V) and LiDAR (12 V), and a boost converter for the stepper drivers (36–40 V). Each converter was selected for high efficiency, ruggedized enclosures, and modular mounting, simplifying maintenance and ensuring resilience to environmental factors. The enclosure design itself balances strength and aesthetics, combining aluminum framing with white-coated steel panels to yield a sleek, professional appearance while facilitating quick access to internal components through removable covers and quick-release fasteners.

In software, a layered architecture on the Jetson Nano cleanly separates configuration, communication, sensing, localization, planning, and execution modules. The Extended Kalman Filter fuses encoder odometry, IMU readings, and optional LiDAR odometry at 100 Hz, guaranteeing precise pose estimation. Dual planners—A* on an 8-connected grid and a sampling-based PRM—allow for dynamic selection between guaranteed shortest paths and smoother “as-the-crow-flies” trajectories. Both online (real-time, low-latency occupancy updates) and offline (loop-closure, global map optimization) SLAM pipelines are implemented, striking a balance between reactive safety and post-mission mapping fidelity. On the lower layer, the ATmega32U4 executes a deterministic 10 ms control loop for motion profiling, sensor parsing, and telemetry streaming—leveraging trapezoidal velocity profiles and Joseph-form EKF updates to ensure both responsiveness and numerical stability.

The remote monitoring and control interfaces provide users with a full-featured, robust PyQt5/pyqtgraph dashboard—complete with live telemetry plots, PID gain tuning, and log panels—as well as a lightweight CLI tool for rapid teleoperation and status monitoring in resource-constrained environments. This dual-interface approach addresses diverse stakeholder needs, from engineers performing system tuning to operators requiring minimal-overhead teleop capabilities.

Safety and serviceability considerations are woven throughout the design: redundant cliff sensors, easily accessible E-stop buttons, and ISO-13482 compliance features ensure safe human-robot interaction. Modular component enclosures, labeled connectors, and clearly documented wiring harnesses minimize downtime during maintenance or upgrades.

In closing, this documentation has laid out a rigorous, data-driven design methodology—from stakeholder analysis and user observations through to detailed hardware, firmware, and software implementations—culminating in a fully integrated AMR prototype. By adhering to clear design principles of modularity, scalability, safety, and

cost-efficiency, we have delivered a robust foundation for both immediate deployment and ongoing innovation in indoor autonomous robotics.